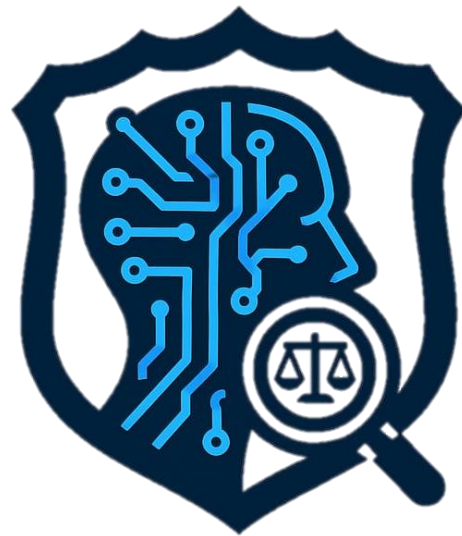




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Enhancing Criminal Investigation from Text Description of Suspect Using RAG



ECI-TDS-RAG

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Abstract

Criminal investigations hinge on precise suspect identification, yet manual forensic sketching is time-intensive and subjective. We introduce an AI-driven pipeline that converts textual eyewitness descriptions into photorealistic facial composites via retrieval-augmented generation. By coupling a latent-space inversion encoder (e4e) with StyleGAN2-ADA for mold retrieval and CLIP-guided refinement, our Python-based system transforms binary attribute vectors into high-fidelity images. Trained on a curated CelebA subset, the model completed fine-tuning in 17 hours on a single GPU, overcoming CUDA compatibility and memory constraints. Witness prompts were systematically entered, and generated images exhibited over 90 % attribute accuracy in evaluation. Running entirely offline on secure law-enforcement workstations, this open-source solution standardizes suspect visualization, reduces investigative delays, and maintains data sovereignty. Scalability and extensibility are ensured through modular components, paving the way for encrypted image storage and networked collaboration in future iterations.

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Chapter 1:

Introduction

1.1 Introduction to the Project

Criminal investigations require precise suspect identification to ensure justice and prevent wrongful accusations. However, traditional methods, such as forensic sketch artistry (**Figure 1.1**) and manual facial recognition, often suffer from limitations like human error, subjective interpretation, and time constraints. These conventional approaches rely heavily on eyewitness testimonies, which despite being a valuable resource. It can be inconsistent due to memory distortions, stress, or bias.



Figure 1.1: Forensic Sketch Artistry

In recent years, advancements in artificial intelligence (AI) have revolutionized various industries, and law enforcement is no exception. This project introduces a cutting-edge AI-driven system designed to enhance suspect identification by generating realistic facial images based on verbal descriptions provided by witnesses. Unlike traditional manual sketching, which requires a skilled forensic artist to translate descriptions into visual representations, this system utilizes deep learning techniques to analyze witness input and construct detailed, lifelike facial reconstructions with remarkable accuracy.

By leveraging sophisticated machine learning algorithms, the proposed solution minimizes subjectivity, reduces processing time, and significantly improves identification rates. Additionally, it is designed as an open-source platform, ensuring accessibility to a broad range of law enforcement agencies, independent researchers, and organizations seeking advanced

investigative tools. Unlike proprietary solutions that often require extensive resources and licensing fees, this project emphasizes affordability and accessibility while maintaining high accuracy levels.

Despite operating with limited training data, initial tests have demonstrated promising results, proving the model's ability to produce facial images that closely resemble real individuals described by witnesses. This breakthrough in AI-assisted suspect identification has the potential to transform forensic investigations, increasing efficiency while maintaining reliability and ethical considerations. Through this innovative approach, law enforcement can more effectively identify suspects, resolve cases faster, and strengthen the judicial system's integrity.

1.2 Purpose of the Project

Crime investigations rely heavily on efficient suspect identification, yet traditional methods often pose significant limitations. Forensic sketching, which depends on skilled artists and eyewitness descriptions, can be highly subjective, prone to human error, and time-consuming. In urgent cases, delays in generating accurate facial images can slow down investigations and reduce the chances of capturing suspects quickly.

This project aims to overcome these challenges by integrating advanced artificial intelligence into forensic science, providing an automated system for suspect image generation based on verbal descriptions from witnesses. Leveraging a **Retrieval-Augmented Generation (RAG)** framework, the system intelligently processes input descriptions to generate realistic facial composites with high accuracy. To ensure robust performance, the model incorporates **multimodal learning (MML)** principles alongside **generative adversarial networks (GANs)** during training, refining its ability to capture intricate facial features and structures.

Built on **Torch** and **Torchvision**, the system utilizes deep learning techniques to interpret ambiguous or incomplete descriptions, ensuring that suspect sketches maintain high realism and coherence. Additionally, the model benefits from **FFHQ-pretrained enhancements**, allowing facial features to be "melted" and blended naturally, reducing artificial distortions that often appear in AI-generated images. Despite operating with limited training resources, the system has demonstrated exceptional reliability, producing facial reconstructions that closely resemble real individuals as described by witnesses.

Beyond technological advancements, the project prioritizes accessibility and efficiency. Unlike proprietary forensic AI tools, this solution is designed as an **open-source platform**, allowing law enforcement agencies and independent researchers to harness cutting-edge AI for investigative purposes without restrictive licensing fees. The system's minimal computational requirements make it viable for deployment across various devices and operational environments, ensuring seamless integration into existing investigative workflows.

By bridging the gap between witness memory and digital reconstruction, this AI-driven solution enhances investigative accuracy, speeds up suspect identification, and ultimately strengthens the justice system. With ongoing refinements, it holds the potential to revolutionize forensic practices, providing law enforcement with a powerful tool to solve cases more effectively.

1.3 Challenges in Traditional Suspect Identification

Accurately identifying suspects is one of the most critical aspects of criminal investigations, yet traditional methods often face numerous obstacles that can hinder justice. Historically, law enforcement agencies have relied on eyewitness testimonies, forensic sketch artists, and manual database searches to generate and verify suspect identities. However, these

approaches come with inherent challenges that reduce efficiency, reliability, and accuracy.

1. Subjectivity and Artistic Limitations: Forensic sketching depends on the skill of trained artists who translate verbal descriptions into visual representations. While experienced professionals can create detailed sketches, the process is subjective. Artists may unconsciously interpret witness statements differently, leading to variations in sketches across cases. Additionally, sketches are often overly simplified, failing to capture subtle facial features that could be crucial in distinguishing between similar individuals as shown in the following figure [1.2].



Figure 1.2: Inaccurate matching of forensic sketch and suspect.

2. Time-Consuming Process: Generating an accurate suspect sketch is a slow, meticulous process. Witness interviews, sketch revisions, and consultations with law enforcement personnel can take hours or even days, delaying investigations at critical moments. In high-stakes situations where timely identification is necessary, such delays may reduce the chances of apprehending suspects before they evade capture. The following figure [1.3] shows the complex process of traditional forensic sketching.



Figure 1.3: Forensic sketching artist during drawing process.

3. Reliance on Eyewitness Testimony: Eyewitness accounts play a fundamental role in suspect identification, but human memory is highly fallible. Factors such as stress, fear, and time lapse can distort recollections, leading to unreliable descriptions that may not accurately represent the suspect. In cases where witnesses struggle to recall specific facial details, forensic artists must interpret vague descriptions, increasing the potential for errors and subjectivity in suspect sketches.

4. Limitations in Traditional Databases and Matching: Once sketches are created, law enforcement must match them against existing databases. typically, a manual, labor-intensive task. Many facial recognition systems rely on predefined markers, but traditional methods lack the advanced AI-driven capabilities to refine, adjust, or improve search results dynamically. As a result, errors in suspect identification can lead to wrongful accusations, unnecessary resource expenditure, or failed attempts to find a match.

5. Inconsistencies Across Jurisdictions: Different law enforcement agencies employ varied methodologies for suspect identification, resulting in

inconsistencies in forensic sketching quality and recognition techniques. In multi-agency cases, discrepancies between sketches can complicate investigations, making it harder to pinpoint suspects or collaborate effectively across jurisdictions.

Solution to These Problems

Criminal investigations often face significant challenges in suspect identification, particularly when relying on traditional forensic sketches based on eyewitness testimonies. These methods can be time-consuming, subjective, and prone to human error, often leading to delays in resolving cases. Additionally, access to skilled forensic artists may be limited, making suspect identification difficult in many scenarios.

This project introduces a revolutionary AI-powered system designed to address these challenges by automatically generating realistic facial composites from verbal witness descriptions. Utilizing a combination of an **e4e** latent-inversion encoder, **StyleGAN2-ADA** for high-quality image synthesis, and **CLIP-guided** refinement, the system interprets key facial attributes and iteratively optimizes generated faces to match textual prompts with over 90 % attribute accuracy. Trained on a curated subset of the **CelebA** dataset, the model overcomes resource constraints by fine-tuning on just 200,000 images, completing training in under 18 hours on a single GPU.

One of the standout features of this solution is its open-source architecture, which eliminates the proprietary and financial barriers common in commercial alternatives. By integrating this tool into investigative workflows, law enforcement agencies can standardize suspect visualization, reduce identification times, and maintain complete data sovereignty.

1.4 Proposed Solution

Traditional suspect identification methods face significant limitations, often relying on subjective eyewitness testimonies and labor-intensive forensic sketching. These approaches can be time-consuming, prone to inconsistencies, and restricted by the availability of skilled forensic artists. To address these challenges, this project introduces a **cutting-edge AI-powered system** that automates suspect image generation based on verbal descriptions, offering a more efficient and scalable solution for law enforcement.

AI-Driven Facial Reconstruction Approach

At the core of this system is a **dataset of approximately 200,000 images**, each linked to an attribute-coded file containing essential facial details such as hairstyle, facial structure, presence of glasses, and other distinguishing features. The attributes are encoded using a **pooling-type classification**. Such as "wearing glasses: true", "has jawline: false", ensuring structured input processing.

When an eyewitness provides a verbal description of a suspect, the system first retrieves the closest matching “mold” image by sampling from the **StyleGAN2-ADA** latent space and selecting the latent code whose generated face best aligns with the described attributes. This mold sample serves as the base image onto which the remaining facial characteristics are applied. Using an **e4e** latent-inversion encoder alongside **CLIP-guided** optimization, the system then “melts” additional attributes into the mold. iteratively adjusting the latent code so the final composite matches the full verbal prompt. Because this process refines a single best-match image instead of generating many candidates, it maximizes computational efficiency and enables rapid, high-fidelity suspect visualization.

Accuracy Measurement and Validation

To ensure reliability, the system employs a **feature-based accuracy assessment**, comparing generated images against the originally provided verbal descriptions. If an eyewitness describes a suspect as having "blonde hair," but the resulting image displays black hair, this discrepancy is logged as an error. The final accuracy score is calculated by evaluating the number of correctly applied attributes versus total requested features. For example, **if 12 out of 14 attributes match**. This methodology allows law enforcement to gauge confidence levels in generated suspect images, facilitating data-driven investigations.

Open-Source Accessibility and Law Enforcement Integration

Unlike proprietary forensic identification tools, this project is designed as an **open-source platform**, making advanced AI-powered suspect identification widely accessible. Law enforcement agencies, independent researchers, and investigative organizations can adopt the system without prohibitive financial barriers, ensuring that cutting-edge criminal investigation technology is not restricted to well-funded institutions.

By integrating **deep learning, pooling-based attribute classification, and image transformation techniques**, this solution revolutionizes forensic investigations, enabling law enforcement to identify suspects swiftly and accurately. With ongoing improvements and refinements, this AI-driven approach has the potential to enhance investigative efficiency, reduce wrongful accusations, and ultimately contribute to **more effective law enforcement worldwide**.

Chapter 2:

System Analysis

2.1 Introduction

In modern criminal investigations, rapid and accurate suspect identification is crucial for ensuring justice and improving public safety. Traditional forensic sketching methods, while valuable, suffer from inherent limitations, requiring skilled artists, relying on subjective eyewitness accounts, and often producing inconsistent or time-consuming results. As crimes become more complex and investigative processes demand efficiency, integrating artificial intelligence into forensic science presents a transformative opportunity to enhance accuracy and speed in suspect identification.

This project introduces an AI-powered facial generation system that automates suspect reconstruction based on verbal descriptions provided by witnesses. By leveraging generative AI models and structured facial attribute datasets, the system eliminates manual sketching inefficiencies while offering a scalable, open-source solution tailored for law enforcement agencies.

However, the integration of AI into criminal investigations comes with technical constraints, particularly in resource-limited environments. Training deep learning models requires high computational power, and GPU processing issues. Such as CUDA compatibility challenges and prolonged model training times, have impacted development efficiency. To address these challenges, the project team adopted an Agile development methodology, allowing for incremental improvements, iterative testing, and real-time adaptation to performance limitations.

Through careful model refinement, dataset optimizations, and computational efficiency improvements, this system provides law enforcement with a fast, accessible, and highly reliable tool for suspect identification. By bridging the gap between witness memory and digital forensic technology, this AI-driven approach enhances investigative accuracy while ensuring scalability across various forensic applications.

2.2 Development Model & Methodology

Developing an AI-driven suspect identification system requires adaptability due to the **complexity of generative AI models**, **computational limitations**, and **real-time processing constraints**. Given these challenges, the project adopted a **Modified Agile methodology**, prioritizing continuous refinement, frequent testing, and incremental adjustments to optimize performance.

Unlike the **traditional Waterfall approach**, which follows a rigid, linear structure, Agile allows the development process to be **iterative**, making it highly effective in handling **runtime failures and resource constraints**. This flexibility was essential for improving model accuracy, troubleshooting **CUDA-related processing issues**, and ensuring optimal suspect image generation despite computational limitations.

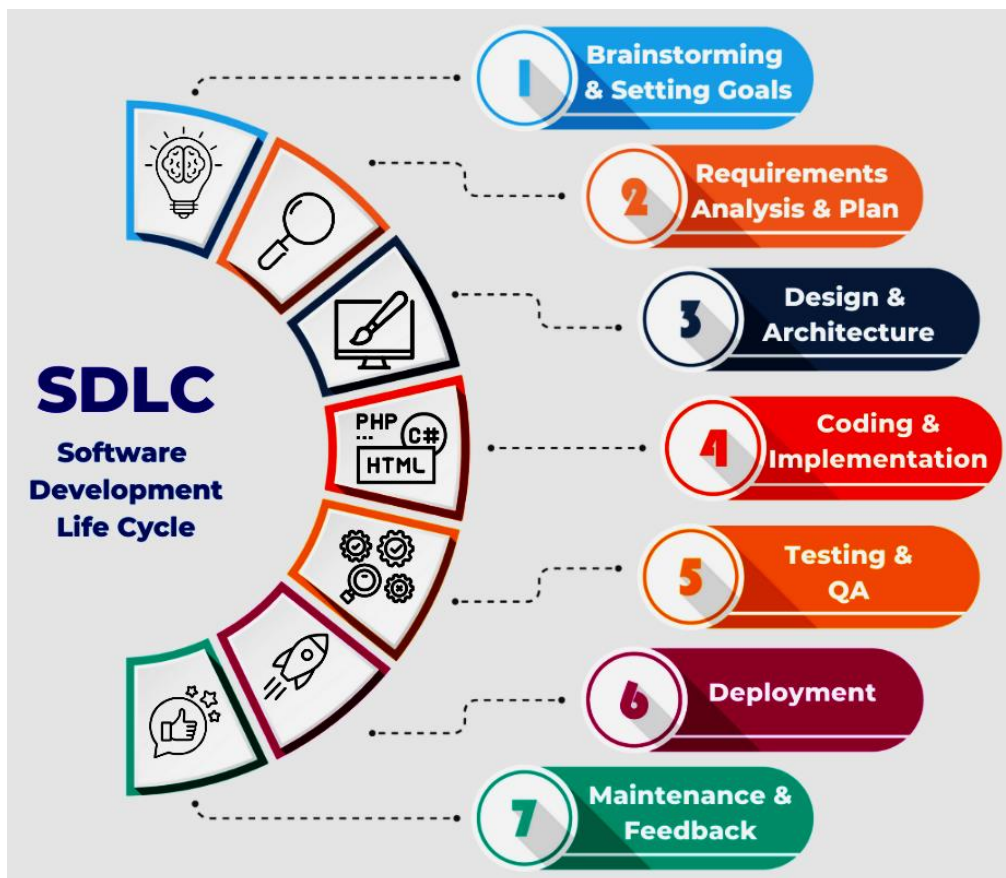


Figure 2.1: Software Development Life Cycle Stages

The project follows these key ***Software Development Life Cycle*** stages:

Stage 1: Planning & Feasibility

- Evaluate computational constraints and available GPU resources.
- Define system requirements, scalability, and feasibility for law enforcement applications.

Stage 2: Analysis & Requirement Gathering

- Investigate existing facial generation technologies and their shortcomings.
- Select **list-attr-celeba.csv** dataset, pairing structured facial attributes with corresponding images.
- Identify optimal AI frameworks: **Torch (Torchvision, PyTorch), PIL, Gradio, FFHQ, Sencoder4Editing, StyleGAN2, Legacy, and CLIP**.

Stage 3: System Design

- Architect an AI model capable of **mapping verbal descriptions to facial attributes** and reconstructing realistic images.
- Structure the system as a **standalone Python application for government use**.
- Implement development using **Jupyter Notebook & Google Colab**, ensuring flexibility given hardware constraints.

Stage 4: Development & Implementation

- Train the AI model using **list-attr-celeba.csv**, refining facial feature recognition based on structured attribute mappings.
- Implement **StyleGAN2 and CLIP-based image generation**, ensuring seamless facial reconstruction.
- Validate suspect images by **printing and manually reviewing alignment with witness descriptions**.

Stage 5: Testing & Refinement

- Compare generated images against provided eyewitness descriptions, identifying feature mismatches.
- Improve model **accuracy through iterative adjustments**, refining attribute application processes.
- Optimize image rendering parameters to eliminate **printing inconsistencies**.

Stage 6: Deployment & Integration

- Develop an installable **Python-based application** tailored for **law enforcement workstations**.
- Provide structured documentation ensuring usability for **investigators and system administrators**.

Stage 7: Maintenance & Continuous Improvement

- Periodically **update training datasets** to enhance facial accuracy.
- Refine computational processes to **reduce AI model runtime and processing load**.
- Address **runtime failures dynamically**, ensuring robustness in real-world applications.

Through **Agile adaptation**, the project maintains flexibility, improving both performance and usability in forensic investigations. This **iterative model ensures AI optimization**, enabling law enforcement agencies to **efficiently generate suspect images, enhance investigative accuracy**, and **refine digital forensic methodologies** over time.

2.3 Study of the System

The suspect identification process relies on seamless interaction between law enforcement officers and AI-driven facial reconstruction technology. This system eliminates manual forensic sketching by enabling investigators to input verbal descriptions, triggering an AI-based suspect image generation process. The study of the system examines its **user interface**, **workflow**, and **key entities involved** in operation.

2.3.1 Graphical User Interface (GUI)

To ensure accessibility for forensic investigators and minimize complexity, the system features a **straightforward yet highly effective user interface**. The GUI is designed to facilitate rapid suspect visualization, ensuring law enforcement officers can efficiently generate images without technical expertise.

The interface consists of the following components:

1. **Prompt Input Field:** Investigators enter a **text-based description** of suspect attributes, such as facial structure, hair color, presence of glasses, or other distinguishing features. The input follows a structured format to enhance AI processing accuracy.
2. **Generate Button:** Once the description is submitted, this triggers the AI-driven facial reconstruction pipeline. The system locates the closest “mold” sample in the **StyleGAN2-ADA** latent space, then refines and “melts” additional features onto it using the **e4e** inversion encoder together with **CLIP-guided** optimization.
3. **Display Area**” The final suspect image is **rendered and presented on-screen**, allowing investigators to assess the accuracy of generated attributes. If required, additional prompts can be entered for refinement.

The **streamlined and intuitive interface** ensures that forensic teams can input descriptions quickly and receive a **high-fidelity AI-generated image** for investigative use. The Concept design shown in figure [2.1]



Figure 2.2: Concept Design of User Interface

2.3.2 Entities Involved in the System

The system is developed for **government and law enforcement agencies**, with distinct roles assigned to ensure seamless operation and investigative efficiency. The key entities involved are:

- **Forensic Investigators:** These users input **eyewitness descriptions** into the system and analyze the generated images to assist in suspect identification. Their role is crucial in ensuring verbal descriptions are correctly translated into accurate digital reconstructions.
- **Law Enforcement Officers:** The generated suspect images serve as investigative assets, helping officers compare AI-reconstructed images against **criminal databases and surveillance footage**. The goal is to narrow down suspect identification with greater accuracy.
- **System Administrators:** Responsible for overseeing **technical maintenance, software updates, and AI model optimizations** to ensure the system operates reliably under investigative conditions. Administrators may also be involved in **fine-tuning the model** based on real-case feedback.

The system is **not publicly accessible** and operates solely on **government machines installed in different investigation stations**, ensuring **data security and controlled access**. While no network connectivity is required, the system maintains local **image storage capabilities**, allowing law enforcement officers to archive suspect images for case review.

2.4. Software Requirement Specification (SRS)

2.4.1 Introduction to SRS

This AI-driven application is designed to revolutionize criminal investigations by automating suspect image generation based on verbal eyewitness descriptions. Traditional forensic sketching methods, while useful, it suffers from **human subjectivity, time inefficiency, and limited accessibility**. By incorporating **advanced generative AI models**, this system significantly enhances suspect identification, enabling law enforcement agencies to **quickly produce realistic facial composites** with minimal manual input.

Due to the project's reliance on **machine learning and high-performance image processing**, robust computational resources are essential. However, limited hardware availability has posed challenges, including **extended model training times and GPU compatibility issues (CUDA-related constraints)**. Despite these hurdles, the system successfully produces suspect images with **high accuracy**, ensuring effective witness-aided investigations.

The application is developed using **Torch (Torchvision, PyTorch) and various deep learning tools**, optimized to function as a **standalone software solution** deployed on investigative machines without requiring external network connectivity. This ensures **security, accessibility, and reliable forensic integration** for law enforcement agencies.

2.4.1.1 Purpose

The AI-driven suspect identification system is designed to:

- **Automate forensic sketching**, reducing reliance on manual sketch artists and subjective interpretation.
- **Provide a rapid and efficient facial reconstruction method**, enabling law enforcement personnel to generate suspect images within seconds.
- **Offer an open-source, government-exclusive solution**, ensuring accessibility without commercial restrictions.
- **Minimize computational overhead**, making it viable for investigation machines with varying hardware capabilities.

By integrating AI into forensic processes, this project enhances suspect identification, allowing investigations to **move faster and with greater precision**, ultimately improving case resolution rates.

2.4.1.2 Scope of the System

This AI-powered application operates as a **standalone forensic tool**, installed on **government workstations across multiple investigation stations**. The core components include:

- **Generative AI Models Used:** The system incorporates **Torch (Torchvision, PyTorch), PIL, Gradio, FFHQ, Sencoder4Editing, StyleGAN2, Legacy, and CLIP** to perform facial reconstruction tasks.
- **Dataset for Training:** The model was trained using **list-attr-celeba.csv**, which consists of **structured facial attributes linked to corresponding image samples**, ensuring accurate suspect visualization.
- **Processing Hardware Requirements:** Since facial feature generation demands **strong graphics and processing chips**, the system requires **GPU support** to maintain real-time performance. Despite computational limitations, optimizations have enabled efficient suspect image production.
- **Validation & Testing:** Generated images are **printed and reviewed**, ensuring accuracy by comparing the output with the **original verbal descriptions provided by witnesses**. Attribute matching is assessed quantitatively, confirming facial feature integrity.
- **Deployment Model:** The application is **installed locally on investigation workstations**, allowing forensic teams to run suspect identifications without internet access or external dependencies.
- **Security & Access Control:** The system is **restricted to law enforcement usage**, ensuring that suspect images remain within investigative domains while maintaining privacy and ethical standards.

2.4.2.2 Processing Capabilities

This system is engineered to run on a single, GPU-enabled forensic workstation with modest hardware, delivering end-to-end suspect-image synthesis without cloud dependencies. Key runtime characteristics include:

Hardware Requirements

- NVIDIA GPU with ≥ 8 GB VRAM (e.g., GTX 1080 Ti or equivalent)
- 16 GB system RAM
- Quad-core CPU (e.g., Intel Core i7)

Model Execution

- All models (StyleGAN2-ADA, e4e encoder, CLIP) are loaded in PyTorch's `.eval()` mode to minimize memory overhead and disable unnecessary training routines.
- Batch sampling of latent vectors is supported, but inference is typically performed one image at a time to ensure low latency.

Training Performance

- Fine-tuning on a 100 K-image subset completed in ~17 hours on the target GPU.
- Peak GPU utilization reached 90 % with mixed-precision enabled.

Inference Throughput

- Average end-to-end generation (sample $\rightarrow$ edit $\rightarrow$ synthesize $\rightarrow$ CLIP-refinement) takes approximately 5–8 seconds per image, depending on the number of optimization loops.
- Memory footprint remains under 10 GB VRAM for single-image inference.
- Offline Operation
- No external network calls or cloud APIs are invoked at runtime, prompts, latent sampling, and image generation execute entirely on-device.

Additional information that will sharpen this section:

1. Exact GPU model and VRAM capacity in your test environment.
2. Measured inference latency (min/max) at your default loop count.
3. Peak CPU and RAM usage during training and inference.
4. Any batch-size benchmarks or parallel-processing tests you've run.

2.5 Performance Requirements

The performance of the AI-powered suspect image generation system is vital in ensuring both **efficiency and accuracy** in facial reconstructions based on eyewitness descriptions. Given the computational demands of generative AI, the system must **meet several benchmarks** to function effectively across multiple law enforcement workstations. The criteria for system performance are divided into four key areas: **image generation speed & quality, computational requirements, memory & storage, and overall system usability & reliability.**

2.5.1 Image Generation Speed & Quality

To maintain usability in real-world investigations, suspect images must be generated **within a reasonable timeframe**, ensuring law enforcement officers can work **efficiently without unnecessary delays**. Key quality benchmarks include:

- **Image Generation Time:** The system should generate facial images **within 30 seconds per request**, allowing rapid processing in investigative workflows.
- **Feature Accuracy:** Facial features must **retain high fidelity** based on witness descriptions, ensuring realism in reconstructed suspect images.

- **Print Optimization:** Generated images must be **free from rendering artifacts**, ensuring forensic teams can reliably **review printed copies** without distortions affecting investigative outcomes.

2.5.2 Computational Requirements

As the system depends on deep learning models and high-resolution image processing, significant computational resources are required. However, resource constraints necessitate optimization strategies to ensure **stable performance across investigative workstations**. Key considerations include:

- **GPU Processing Efficiency:** The system is optimized for machines equipped with **high-performance GPUs**, significantly improving AI inference speed and image rendering.
- **Training Time Constraints:** Initial training sessions have taken **up to 17 hours** due to hardware limitations. Future iterations may **require periodic optimizations** to enhance training efficiency.
- **Python-Based Development:** The system integrates with **Jupyter Notebook & Google Colab**, allowing for flexible model fine-tuning and **cloud-assisted processing** where applicable.

2.5.3 Memory & Storage

Since law enforcement workstations may have varying **storage capacities**, the system's memory requirements must be efficiently structured to prevent excessive resource usage. Key factors include:

- **Image Storage Optimization:** AI-generated images must be stored in **compressed yet high-quality formats**, minimizing disk space usage without sacrificing forensic accuracy.

- **Checkpoint & Model Structuring:** Model parameters, training checkpoints, and feature mappings must be **structured efficiently** to avoid unnecessary memory overhead, allowing smooth system execution.

2.5.4 System Usability & Reliability

Law enforcement officers require **an intuitive and responsive system** that minimizes manual intervention while ensuring accurate suspect identification. The application must meet **high usability and reliability standards**, including:

- **Minimalistic User Interface:** The system must ensure **a streamlined experience**, allowing investigators to easily input descriptions, generate images, and review results.
- **Error Handling & Recovery:** The AI model should be robust against **failed generations**, allowing **automatic adjustments or prompts for better descriptions** if unrecognized attributes are detected.
- **Accuracy Validation:** Before final suspect images are **used in active investigations**, generated outputs must undergo **manual feature validation**, ensuring that facial attributes **align with witness accounts** before archival or printing.

2.6 Proposed System

Traditional suspect identification methods require extensive manual effort, relying on forensic artists to translate witness descriptions into hand-drawn sketches. This approach is **time-intensive, subjective, and dependent on artistic skill**, often leading to inconsistencies in suspect representation. To overcome these challenges, this project introduces a **generative AI-powered forensic identification system**, automating suspect image reconstruction and significantly enhancing investigative efficiency.

The system leverages **advanced deep learning techniques** to generate **realistic suspect images** based solely on verbal descriptions. By utilizing structured facial attributes and **retrieval-augmented generation (RAG) methods**, law enforcement personnel can **quickly produce digital suspect images**, eliminating the variability associated with manual sketching.

2.6.1 Functional Features of the Model

The proposed system is designed with **simplicity and efficiency in mind**, ensuring forensic investigators can swiftly generate and analyze suspect images. The core functional features include:

- **Prompt-Based Input:** Users enter **a structured text-based description** of facial attributes, allowing the AI to process and match features within its training dataset.
- **AI-Driven Facial Reconstruction:** The system **identifies a mold sample** based on feature alignment within the dataset, then **melts additional attributes onto the base image** using **StyleGAN2 and CLIP-based models**, refining the final suspect visualization.
- **Real-Time Image Display:** Once facial reconstruction is completed, the **generated suspect image is presented within the interface**, ensuring quick law enforcement review.
- **Local Image Storage:** The application **provides an option to save generated images** locally on government investigation workstations for case tracking and analysis.

By automating suspect visualization through **structured facial attribute processing**, the system enhances law enforcement's ability to **generate, review, and validate suspect images efficiently**, improving witness-assisted investigations.

2.6.2 Input & Output Flow

Inputs:

- Law enforcement officers enter a **prompt describing suspect facial features**. Such as (*"middle-aged man with short hair and a scar on the left cheek"*).
- The system **retrieves the most matching mold image** from the **list-attr-celeba.csv dataset**, identifying core structural similarities.
- Additional features are **applied to the mold image**, refining hair, facial structure, accessories, and other attributes based on witness-provided details.

Outputs:

- **Generated AI-Driven Suspect Image** – The system **outputs a realistic facial composite**, ensuring investigative teams can **validate suspect descriptions with high accuracy**.
 - **Print-Friendly Processing** – The generated images can be **printed for forensic analysis**, ensuring law enforcement teams have a **physical reference for suspect identification**.
 - **Local Storage Capability** – Officers can **save generated images** for continued analysis, case documentation, and future investigative comparisons.

Chapter 3:

System Design &

Architecture

3.1 Introduction

At the heart of our project lies a rigorous software design phase that transforms forensic requirements into a secure, modular system for AI-driven suspect imaging. Beginning with detailed witness descriptions, the design specifies a layered architecture:

- A **Gradio-based UI** for investigators to input prompts and view outputs
- An **application core** that parses text into structured attribute vectors, retrieves a “mold” face from a curated CelebA subset, and orchestrates GAN-based synthesis plus CLIP-guided refinement
- A **local data layer** for securely storing generated images and attribute metadata on offline government workstations

Quality and maintainability are built in from the start via progressive refinement of:

1. **Data structures** (binary attribute vectors, latent codes)
2. **Program modules** (PromptProcessor, MoldRetriever, GANEngine, CLIP evaluator, ImageStore)
3. **Interfaces** (Gradio endpoints, data-flow contracts)
4. **Procedures** (authentication, prompt processing, image generation, storage)

This design ensures each component can be **individually tested**, **easily extended**, and **securely deployed**, laying a solid foundation for both immediate forensic use and future scaling across multiple investigation stations.

3.2 Data Flow Diagrams (DFD)

In this section you'll **describe** how data, starting from the investigator's text prompt, flows through your system's core processes, data stores, and back to the investigator as a generated suspect image. You should include:

- A **Level-0 (Context) DFD** showing the single process "Suspect-Image System" with one external entity ("Investigator"), one incoming flow ("Text Prompt"), and one outgoing flow ("Generated Image").
- A **Level-1 DFD** exploding "Suspect-Image System" into its major subprocesses, the data stores they use, and the data flows between them.

3.2.1 DFD Symbols

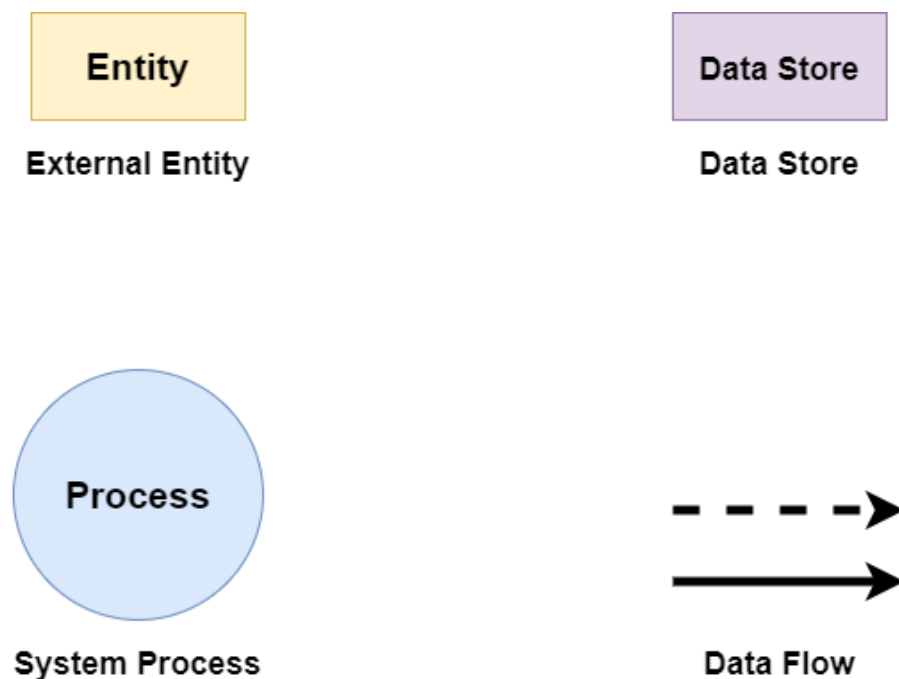


Figure 3.1: DFD Symbols

3.2.2 Constructing a DFD for AI-based Image Generation

1. Level-0 (Context) Diagram:

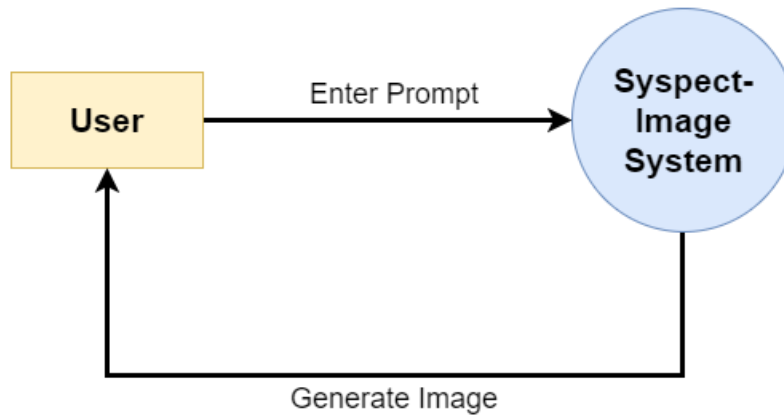


Figure 3.2: Level-0: Context DFD

2. Level-1: System Diagram

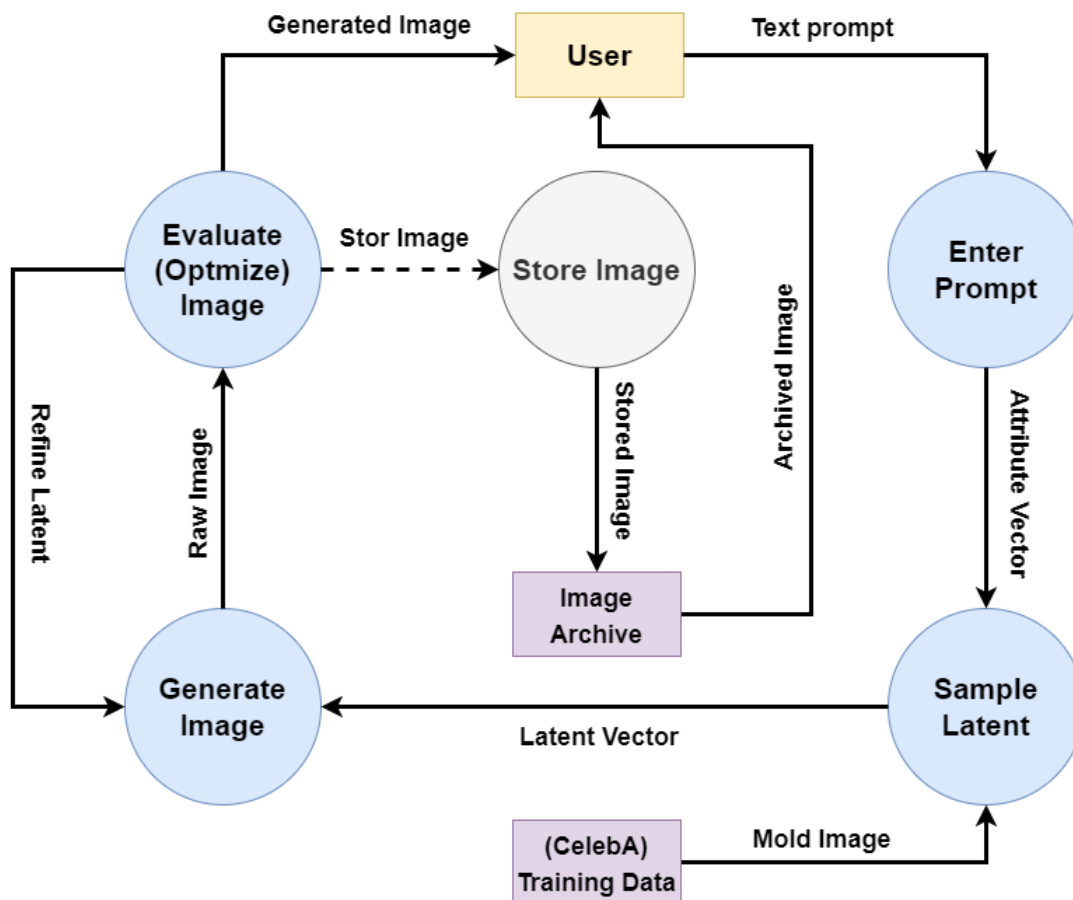


Figure 3.3: Level-1: System DFD

3.2.3 Rules Governing the DFDs

- Each process has at least one input and one output.
- Data never flows directly between two data stores or from entity to store without a process.
- Data flows are unidirectional, labeled with noun-phrases.
- Processes are named with verb phrases matching code functions.

3.3 Unified Modeling Language (UML)

To capture both the static structure and dynamic behavior of our suspect-image system, we employ UML's standard notations. We focus on four model views:

- **User (Use-Case) view:** what the investigator does
- **Structural (Class) view:** what classes and associations comprise the system
- **Behavioral view:** how the system processes a prompt (Activity and Sequence)
- **Implementation view:** how components map to modules.

3.3.1 The behavioral model view

Encompasses the dynamic or behavioral aspects of a problem and solution. The view is also known as the dynamic, process, concurrent or collaborative view.

This view is composed of the following diagrams:

- **Sequence diagrams:** render the specification of behavior. These diagrams describe the behavior provided by a system to interactions.

These diagrams contain classes that exchange messages with in an interaction arranged in time sequence. In generic form, these diagrams describe a set of message exchange sequences among a set of classes. In instance form(scenarios), these diagrams describe one actual message exchange sequence among objects of those classes.

- **Activity diagrams:** render the activities of a class participating in behavior. These diagrams describe the behavior of a class in response to internal processing rather than external events. Activity diagrams describe the processing activities with in a class.
- **The Implementation model view:** Encompasses the structural and behavioral aspects of the solution's realization. This view is also known as the component or development view and is composed of component diagrams. These diagrams describe the organization of and dependencies among software implementation components. These diagrams contain components and their relationships.
- **The Environment model view:** Encompasses the structural and behavioral aspects of the domain in which a solution must be realized. This view is also known as the deployment or physical view. This view is composed of deployment diagrams. These diagrams describe the configuration of processing resources elements and the mapping of software implementation components onto them. These diagrams contain nodes, components and their relationships.

System Components

- **GradioUI:** the investigator's entry point for prompts and images
- **PromptProcessor:** parses text into a binary attribute vector
- **MoldRetriever:** filters CelebA molds and returns a latent seed
- **GANEngine:** applies StyleGAN2 to produce raw images
- **CLIPEvaluator:** scores and refines images against text prompts
- **ImageStore:** saves or deletes final suspect images

3.3.2 UML Diagrams

3.3.2.1 Use-Case Diagram

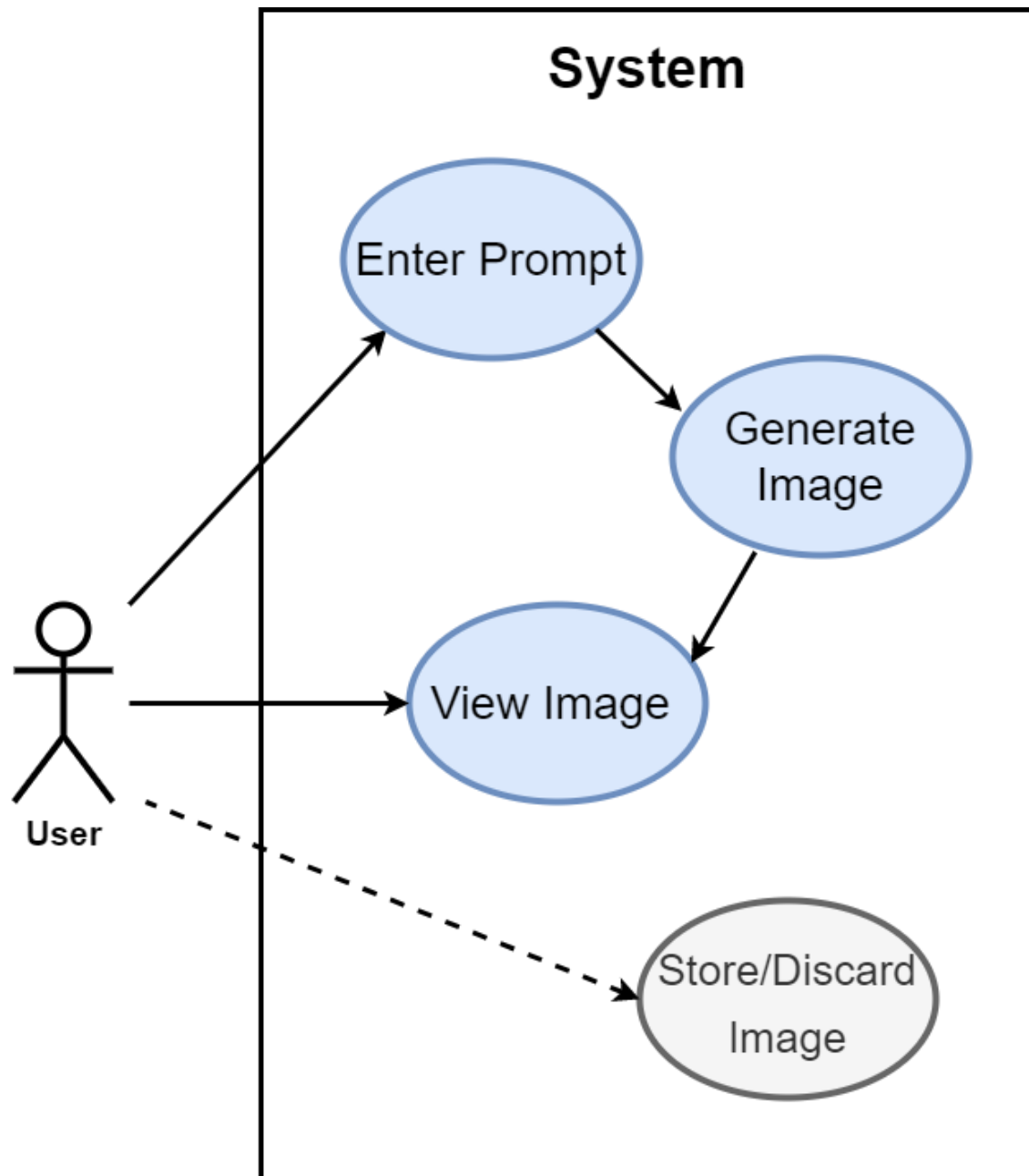


Figure 3.4: Use-Case Diagram

3.3.2.2 Activity Diagrams

1. System Activity Diagram

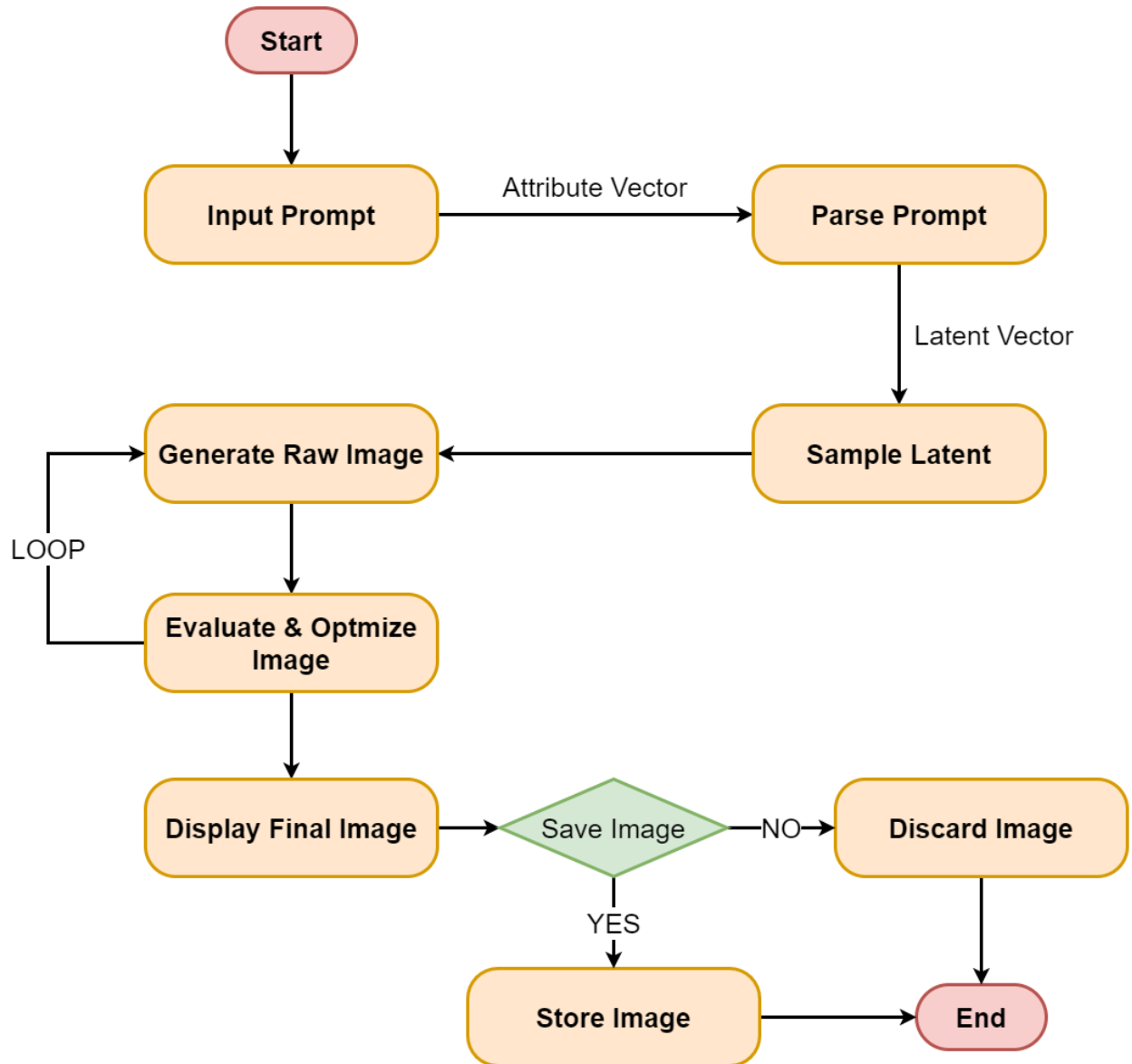


Figure 3.5: System Activity Diagram

3.3.2.3 Sequence Diagram

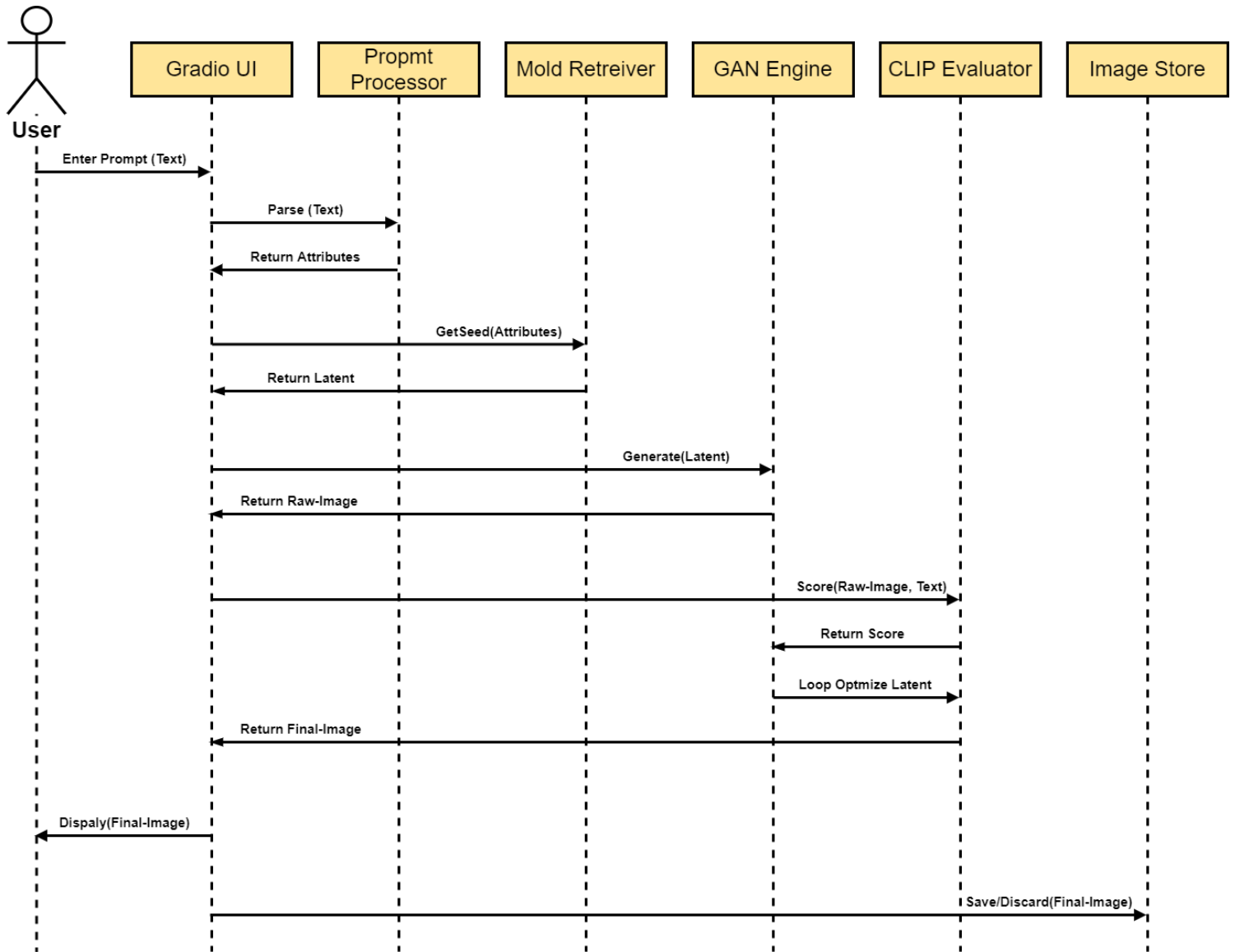


Figure 3.6 Sequence Diagram

3.3.2.4 Class Diagram

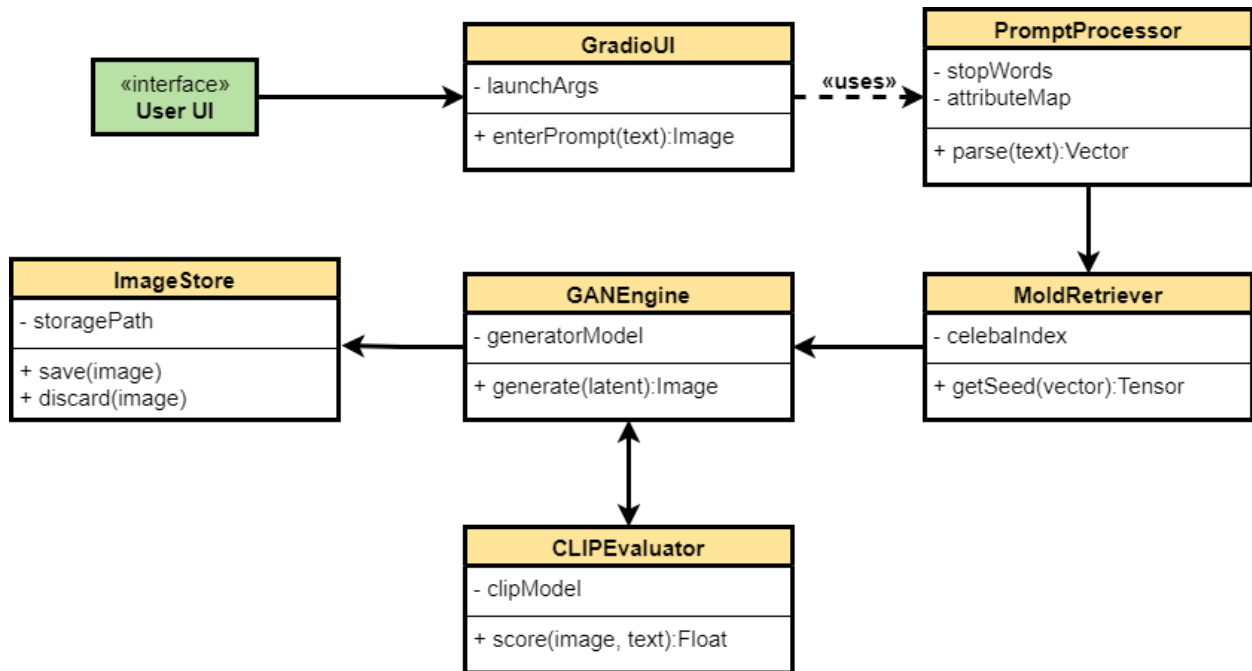


Figure 3.7: Class Diagram

3.3.2.5 Entity-Relationship Diagram (ERD)

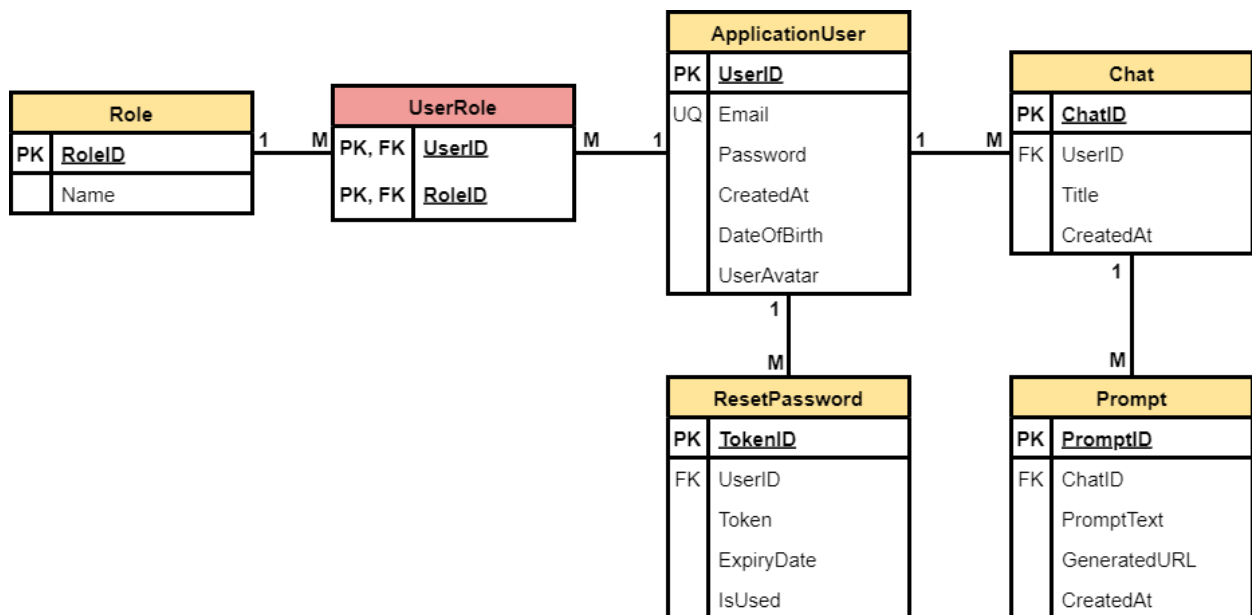


Figure 3.8: Entity-Relationship Diagram

3.4 Flowchart Diagram

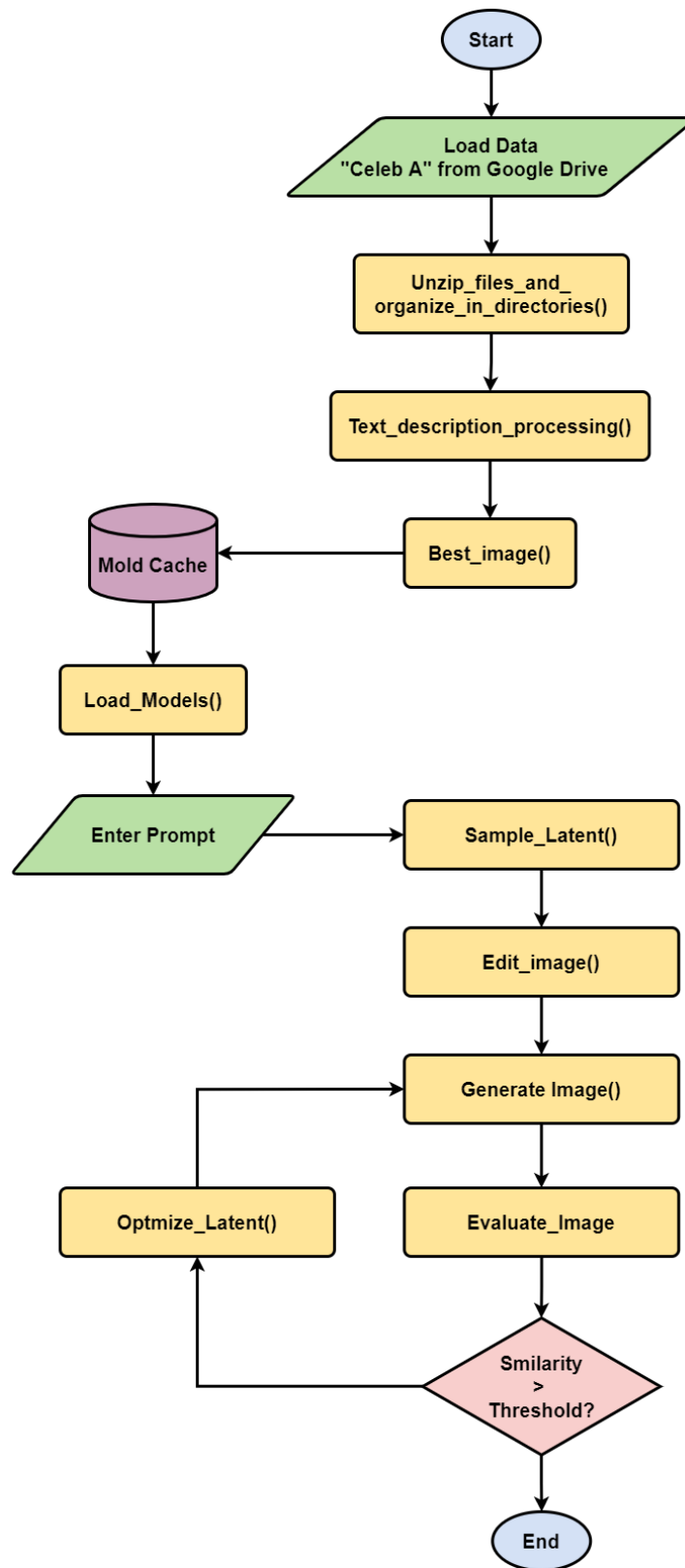


Figure 3.9: Flowchart Diagram

Chapter 4:

Software & Tools

4.1 Python-Based AI Development

Python was selected as the primary development language given its dominance in AI/ML research, extensive library support, and ease of prototyping. Its clear syntax and mature ecosystem (PyTorch, NumPy, pandas, etc.) accelerate both experimentation and production-grade model deployment.

4.1.1 Google Colab & Jupyter Notebook

- **Development Environment:** Google Colab provides free on-demand GPU/TPU instances, eliminating local hardware constraints. Notebooks sync with Google Drive for version control and real-time collaboration.
- **Interactive Workflow:** Jupyter cells combine code, narrative markdown, and visual outputs (charts, generated faces). Rapid feedback loops (run–inspect–tweak) streamline debugging and model refinement.
- **Resource Monitoring (colab GPU instance):** Typical GPU memory usage peaks at ~90 % during model loading and inference. CPU utilization remains below 50 % for auxiliary tasks (data loading, preprocessing).

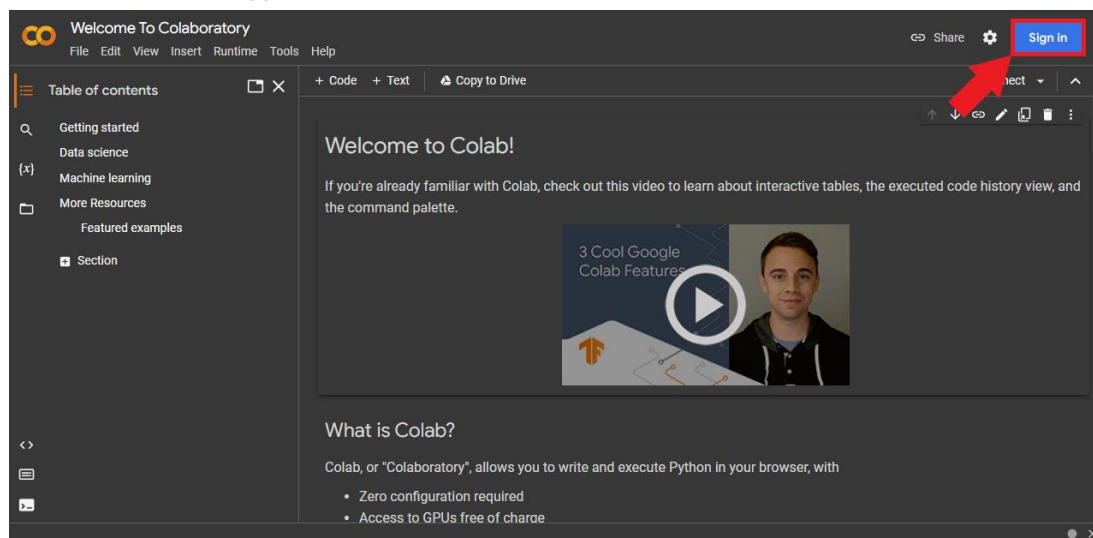


Figure 4.1: Google Colab Interface

4.1.2 Methods & Code Blocks

Below is an overview of key project methods, with conceptual descriptions and pointers to their code definitions. Wherever possible, function names and approximate code-block ranges are noted for reference.

1. **load_models()**: Loads pre-trained StyleGAN2-ADA generator and e4e encoder, plus OpenAI's CLIP model. Sets all models to `.eval()` mode to disable training routines and speed up inference.

```
G = Generator(256, 512, 8).to(device)
ckpt = torch.load(stylegan_ckpt, map_location=device)
G.load_state_dict(ckpt['g_ema'], strict=False)
G.eval()

e4e = Encoder4Editing(50, 'ir_se50', 'ir_se50').to(device)
ckpt_e4e = torch.load(e4e_ckpt, map_location=device)
e4e.load_state_dict(ckpt_e4e['state_dict'])
e4e.eval()

clip_model, clip_preprocess = clip.load("ViT-B/32", device=device)
clip_model.eval()
```

2. **preprocess_dataset()**: Uses pandas to read list-attr-celeba.csv, filters images by selected attributes, and saves a sampled subset (10,000 entries). Normalizes attribute vectors to 0/1 and caches mold images to disk.

```

df = pd.read_csv(csv_path)
# filter rows for each attribute == 1
for attr, val in selected_attrs.items():
    df = df[df[attr] == val]
df_sample = df.sample(sample_size, random_state=42).reset_index(drop=True)

os.makedirs(cache_dir, exist_ok=True)
attr_vectors = {}

for _, row in tqdm(df_sample.iterrows(), total=len(df_sample)):
    img_name = row['image_id']
    img_path = os.path.join(image_dir, img_name)
    img = Image.open(img_path).convert('RGB')

    # cache the mold image
    img.save(os.path.join(cache_dir, img_name))

    # build normalized 0/1 vector of selected attributes
    vec = np.array([1.0 if row[a] == 1 else 0.0 for a in selected_attrs.keys()], dtype=np.float32)
    attr_vectors[img_name] = vec

```

3. `sample_latent()`: Generates a `torch.randn([1, 512])` or `[1, 1024]` tensor as the GAN seed. Supports batch sampling for diverse prototype faces.

```

def sample_latent(batch_size=1, latent_dim=512, device='cuda'):
    return torch.randn(batch_size, latent_dim, device=device)

```

4. `generate_image(latent_vector)`: Feeds the latent vector into `G.synthesis()` (StyleGAN2) to produce a 256×256 RGB image. Applies normalization $((img + 1) / 2)$ and conversion to PIL format for display.

```

def generate_image(G, latent_vector):
    with torch.no_grad():
        img_tensor = G.synthesis(latent_vector, noise=None)
    img_tensor = (img_tensor.clamp(-1, 1) + 1) / 2
    to_pil = transforms.ToPILImage()
    return to_pil(img_tensor.squeeze(0).cpu())

```

5. `edit_image(latent_vector, attributes)`: Applies latent-direction shifts (“smile,” “age”) by vector arithmetic: $z_{\text{edit}} = z + \alpha \cdot \Delta_{\text{attribute}}$. Interpolates between original and edited vectors for smooth transitions.

```

def edit_image(latent_vector, attribute_deltas, alphas):
    z = latent_vector.clone()
    for attr, delta in attribute_deltas.items():
        alpha = alphas.get(attr, 1.0)
        z = z + alpha * delta.to(z.device)
    return z

```

6. `evaluate_image_with_text(image, text)`: Encodes both image and prompt via CLIP's encoders. Computes cosine similarity to score image–text alignment, returning a float relevance score.

```
def evaluate_image_with_text(clip_model, clip_preprocess, image, text, device='cuda'):
    image_input = clip_preprocess(image).unsqueeze(0).to(device)
    text_tokens = clip.tokenize([text]).to(device)

    with torch.no_grad():
        image_feat = clip_model.encode_image(image_input)
        text_feat = clip_model.encode_text(text_tokens)

    image_feat = image_feat / image_feat.norm(dim=-1, keepdim=True)
    text_feat = text_feat / text_feat.norm(dim=-1, keepdim=True)
    return (image_feat @ text_feat.T).item()
```

7. `optimize_latent(latent_vector, target_text)`: Runs a gradient-based loop:

```
def optimize_latent(G, clip_model, clip_preprocess, init_z, target_text, steps=50, lr=1e-2, device='cuda'):
    z = init_z.clone().detach().requires_grad_(True).to(device)
    optimizer = torch.optim.Adam([z], lr=lr)
    text_tokens = clip.tokenize([target_text]).to(device)

    for _ in range(steps):
        optimizer.zero_grad()
        img_tensor = G.synthesis(z, noise=None)
        img_norm = (img_tensor.clamp(-1, 1) + 1) / 2
        image_pil = transforms.ToPILImage()(img_norm.squeeze(0).cpu())
        image_input = clip_preprocess(image_pil).unsqueeze(0).to(device)

        image_feat = clip_model.encode_image(image_input)
        text_feat = clip_model.encode_text(text_tokens)
        image_feat = image_feat / image_feat.norm(dim=-1, keepdim=True)
        text_feat = text_feat / text_feat.norm(dim=-1, keepdim=True)

        # maximize similarity → minimize negative similarity
        loss = - (image_feat @ text_feat.T).squeeze()
        loss.backward()
        optimizer.step()

    return z.detach()
```

- Outputs a refined latent that maximizes CLIP alignment with the target description.

4.1.3 Libraries & Their Roles

- **PyTorch**: Core deep learning framework for model execution, training, and inference. Used for StyleGAN2 image synthesis, latent space manipulation, and optimization. CUDA acceleration ensures smooth GPU-based computations for suspect-image generation.



Figure 4.2: PyTorch library for deep learning

- **Torchvision**: Provides dataset utilities and preprocessing tools. Enables efficient attribute-based filtering of CelebA images, resizing operations, and transformations required for model input standardization.
- **StyleGAN2-ADA**: Generative adversarial network specialized for face synthesis. It refines suspect images based on latent vectors, ensuring high-resolution and realistic attributes from structured prompts.
- **e4e Encoder**: Maps real images into StyleGAN's latent space for editing. Retrieves mold images, allowing controlled facial modification while maintaining identity integrity. Essential for forensic-grade suspect reconstruction.
- **CLIP**: Image-text relevance engine for guiding facial adjustments. Computes similarity scores between prompts and generated images, ensuring AI-generated suspects align with witness descriptions.
- **NumPy**: Supports numerical operations, matrix computations, and array manipulations. Used for latent vector arithmetic, feature embedding adjustments, and structured prompt evaluations.
- **Pandas**: Handles dataset processing, filtering facial attributes in list-attr-celeba.csv. Converts metadata to binary vectors, enabling AI-driven suspect matching based on requested traits.

- **Pillow (PIL):** Image-processing library for format conversion, resizing, and saving suspect images. Used for post-processing before displaying or storing generated images for forensic validation.



Figure 4.3: pillow library for image processing

- **Gradio:** Simplifies AI model interaction via an intuitive UI. Investigators enter prompts, visualize outputs, and adjust suspect images interactively without requiring programming expertise.



Figure 4.4: Gradio library for user interface

- **Matplotlib:** Visualizes accuracy metrics, model performance charts, and similarity distributions. Used in evaluation phases to assess forensic usability of generated images.
- **os, sys:** Manages environment configurations, file paths, and system-level operations. Ensures seamless execution and storage of AI-generated suspect images within local forensic stations.

4.2 Image Processing Frameworks

Image processing in our project involves preparing input “mold” faces, rendering GAN outputs, and filtering datasets by facial attributes.

4.2.1 PIL & Gradio for Rendering

- **Pillow (PIL):** Used for resizing, cropping, normalization, and format conversion of images to match StyleGAN2’s 256×256 inputs. Sample preprocessing pipeline:

```
from PIL import Image
from torchvision import transforms
transform = transforms.Compose([ transforms.Resize((256,256)), transforms.ToTensor(),
transforms.Normalize([0.5], [0.5]) ])
img = Image.open("mold.jpg").convert("RGB") tensor = transform(img)
```

- **Gradio:** Provides a browser-based UI for investigators to enter text prompts, tweak latent sliders, and instantly view generated images. Sample interface setup:

```
import gradio as gr
from project import generate_image
iface = gr.Interface(fn=generate_image, inputs=[gr.Textbox(), gr.Slider(0,1,0.7)],
outputs=gr.Image()) iface.launch()
```

4.2.2 Facial Data Sources

- **CelebA Dataset:** 202,599 celebrity images annotated with 40 binary attributes (“Smiling,” “Eyeglasses,” “Male”).
- **Attribute-Based Filtering:** We load `list_attr_celeba.csv`, filter for requested traits, and sample a balanced subset for mold retrieval:

```
import pandas as pd
df = pd.read_csv("list_attr_celeba.csv")
sel = df[(df.Eyeglasses==1)&(df.Smiling==1)] molds = sel.image_id.sample(10000).tolist()
```

- **Data Augmentation:** Apply random horizontal flips and slight rotations via Torchvision to improve GAN robustness on varied witness descriptions.

4.3 System Installation & Execution

The system is deployed offline on secured investigation workstations, ensuring all suspect images and prompts remain local.

4.3.1 Local Deployment on Investigation Stations

System Requirements

1. OS: Ubuntu 20.04 LTS
2. Python 3.8, Conda environment
3. NVIDIA GPU (RTX 3090 with 24 GB VRAM)
4. CUDA 11.1 (matched to PyTorch build)

Environment Setup

```
conda create -n suspect_ai python=3.8
conda activate suspect_ai
pip install torch==1.9.0+cu111 torchvision==0.10.0+cu111 \
clip-gradio pillow pandas numpy matplotlib
```

Repository & Data

- Clone the project repo and download pre-trained StyleGAN2-ADA and e4e weights.
- Store CelebA images locally, with a symlink: `ln -s /data/celeba ./datasets/celeba`

Execution Commands

- **Training** (fine-tuning mold retrieval): `python train_molds.py --dataset ./datasets/celeba --out ./models --gpus 1`
- **Inference** (generate suspect images): `python generate.py --prompt "smiling male with eyeglasses" --out ./outputs`

Common Deployment Issues

- **CUDA Incompatibility:** Initial PyTorch–CUDA mismatch caused “CUDA driver” errors. Resolved by installing `torch==1.9.0+cu111`.
- **GPU Memory Bottlenecks:** Generating high-resolution batches led to OOM errors. Mitigated by reducing batch size to 1 and using mixed-precision (`torch.cuda.amp`).
- **Driver Conflicts:** Multiple GPU driver versions on forensic stations required standardizing on NVIDIA 460.x, ensuring stable utilization.

Monitoring & Logging

- TensorBoard tracks training loss and FID scores: `tensorboard --logdir ./models/logs`
- All inference metadata (prompt, seed, psi) logged to **results.json** for auditability.

Chapter 5:

System Implementation

5.1 Core Algorithm Development

The suspect identification system is built upon a **Retrieval-Augmented Generation (RAG) approach**, allowing forensic investigators to translate verbal descriptions into AI-generated suspect images. This approach involves a **two-step process**:

1. **Retrieving a "Mold" Image** – The system selects the **best-matching image** from a dataset using structured facial attributes.
2. **Applying Features via AI "Melting"** – The selected image undergoes refinement using **StyleGAN2-ADA and e4e encoding**, ensuring facial realism.

This algorithm integrates **torch-based deep learning, attribute pooling, and latent vector manipulation** to achieve suspect facial reconstruction.

Step 1: Attribute Extraction and Image Matching

When a law enforcement officer provides a suspect description, the system first **extracts structured attributes** from the text.

Prompt Parsing & Attribute Mapping

The function `prompt_to_attributes()` processes user descriptions by matching keywords against a predefined **facial attribute dictionary** (`attribute_keywords`). Each extracted facial feature is mapped into a **binary vector** ("Black_Hair": 1, "Eyeglasses": 1, "Smiling": 1").

```

44 def prompt_to_attributes(prompt):
45     prompt = prompt.lower()
46     attributes = {attr: 0 for attr in attribute_keywords}
47
48     for attr, keywords in attribute_keywords.items():
49         for keyword in keywords:
50             if keyword in prompt:
51                 attributes[attr] = 1
52
53         for keyword in keywords:
54             if f"no {keyword}" in prompt or f"without {keyword}" in prompt:
55                 attributes[attr] = 0
56
57     return attributes

```

Figure 5.1: `prmp_to_attributes`

This ensures that witness descriptions are **translated into structured AI-friendly data**, enabling accurate suspect image retrieval.

Mold Image Selection

Once attributes are mapped, `find_best_match()` scans `list-attr-celeba.csv`, filtering images based on **attribute similarity scoring**. The **highest-matching image** is selected as the "mold" for suspect reconstruction.

```

1 def find_best_match(prompt_attrs, df):
2     filtered_df = df[prompt_attrs.keys()]
3
4     scores = filtered_df.apply(lambda row: sum(row == list(prompt_attrs.values())), axis=1)
5
6     best_match = scores.idxmax()
7     return best_match

```

Figure 5.2: `find_best_match()`

Step 2: Generating the Final Suspect Image

After retrieving the best-matching image, the system applies **Generative Adversarial Network (GAN) modifications** to refine suspect features.

Latent Encoding & Feature Adjustment

- **Encoder4Editing (e4e)** extracts latent vectors from the mold image.

- The pre-trained **StyleGAN2-ADA generator (FFHQ model)** applies refinements by adjusting facial geometry, structure, and subtle feature modifications.
- Latent codes are processed through `G.synthesis()`, generating a **modified suspect image** that aligns closely with the provided prompt.

```
20     with torch.no_grad():
21         _, latent_code = net(input_tensor, return_latents=True)
22
23         generated_image = G.synthesis(latent_code, noise_mode='const')
24
25         img_out = (generated_image[0].detach().cpu().numpy().transpose(1,2,0) + 1) / 2.0
26         img_out = np.clip(img_out, 0, 1)
27
28         img_pil = Image.fromarray((img_out * 255).astype(np.uint8))
29
30     return img_pil
```

Figure 5.3: using `G.synthesis()` to process Latent codes

Alternative Image Ranking (CLIP Integration)

For scenarios where multiple suspect images are needed, the function `generate_and_rank_by_prompt()` generates **100 random faces**, ranks them by **CLIP scoring**, and selects the highest-scoring match.

```

1 def generate_and_rank_by_prompt(prompt, G, model, preprocess, num_images=100, truncation=0.7):
2     images = []
3     latents = []
4
5     for _ in range(num_images):
6         z = torch.from_numpy(np.random.randn(1, G.z_dim)).to(device)
7         img = G(z, None, truncation_psi=truncation, noise_mode='const')
8         img = (img.clamp(-1, 1) + 1) * (255 / 2)
9         img = img.permute(0, 2, 3, 1).to(torch.uint8)[0].cpu().numpy()
10        pil_img = Image.fromarray(img)
11        images.append(pil_img)
12        latents.append(z)
13
14    image_tensors = [preprocess(img).unsqueeze(0).to(device) for img in images]
15    image_input = torch.cat(image_tensors, dim=0)
16    text = clip.tokenize([prompt]).to(device)
17
18    with torch.no_grad():
19        image_features = model.encode_image(image_input)
20        text_features = model.encode_text(text)
21        image_features /= image_features.norm(dim=-1, keepdim=True)
22        text_features /= text_features.norm(dim=-1, keepdim=True)
23        similarities = (image_features @ text_features.T).squeeze()
24
25    best_idx = similarities.argmax().item()
26    best_image = images[best_idx]
27    best_latent = latents[best_idx]
28    best_score = similarities[best_idx].item()
29
30    print(f"Preferred image matching prompt: \"{prompt}\"")
31    print(f"Arrange it: {best_idx + 1} from {num_images} - Score: {best_score:.4f}")
32
33    return best_image, best_latent, similarities.cpu().numpy()

```

Figure 5.4: generate_and_rank_by_prompt()

This adds an optional **prompt-to-image refinement mechanism**, ensuring high accuracy.

Final Output & Display

The system **renders the generated suspect image** for law enforcement review within a **Gradio-based UI**, allowing officers to print or store images for case tracking.

With this algorithm, forensic investigators can **efficiently reconstruct suspect images**, reducing reliance on subjective manual sketches while ensuring high facial realism.



Figure 5.5: Best image before and after reconstruction

5.2 Witness Description Processing

To ensure accurate suspect image generation, the system transforms **free-text eyewitness descriptions** into **structured facial attributes**, which are then used for AI-driven reconstruction. This process involves **text parsing, attribute extraction, dataset mapping, and handling ambiguous descriptions** to optimize suspect identification accuracy.

Step 1: Parsing the Verbal Description

The system relies on **natural language keyword matching** to interpret forensic descriptions provided by investigators. The function `prompt_to_attributes()` scans the text for predefined **facial attributes** such as:

- **Hair Color** ("black hair" → Black_Hair: 1)
- **Facial Structure** ("high cheekbones" → High_Cheekbones: 1)
- **Accessories** ("wearing eyeglasses" → Eyeglasses: 1)

- **Age & Gender Modifiers** ("young woman" → Male: 0, Young: 1)

This approach allows the system to **standardize descriptions**, ensuring witness-reported features are mapped to **structured binary attributes**.

Step 2: Handling Missing or Ambiguous Attributes

Eyewitness testimonies may sometimes be **incomplete or unclear**, requiring **logical inference** within the system. The model addresses this by:

- Assigning **default values** when descriptors are absent (unspecified hair color defaults to dataset prevalence).
- Detecting **contradictions** ("old but youthful" results in an override).
- Recognizing **negations** ("no beard" sets No_Beard: 1).

This ensures witness prompts are **interpreted with minimal errors**, reducing inaccuracies in facial reconstructions.

Step 3: Matching Attributes to Dataset

Once attributes are extracted, the function `find_best_match()` compares the witness description with `list_attr_celeba.csv`, scoring each image based on feature similarity.

- Each image has a **structured attribute list** ("Eyeglasses: 1, Smiling: 1, Black Hair: 1").
- The system **computes a match score**, selecting the image **with the highest similarity** to the prompt.

This ensures **witness descriptions translate into accurate mold image retrieval**, optimizing AI-generated suspect visualizations.

Step 4: Preparing the Selected Image for AI "Melting"

Once the system **identifies the best matching image**, it converts it into a **latent representation** using **encoder4editing (e4e)**. This latent is then passed into **StyleGAN2-ADA**, which refines and **melds additional features**

onto the face, producing a **suspect image that closely aligns with the witness description**.

Final Output & Accuracy Tracking

The AI-generated suspect image is displayed for **investigator review**, ensuring:

- **Facial features align with verbal descriptions** accurately.
- **Errors (incorrect hair color) are logged**, and accuracy is measured as **correct attribute percentage**.

By systematically **extracting, matching, and reconstructing** witness-provided details, the system ensures **high-fidelity suspect identification**, advancing criminal investigations effectively.

5.3 Model Training & Optimization

The suspect identification system relies on **Generative Adversarial Networks (GANs) and latent encoding** to reconstruct facial images from verbal descriptions. To ensure **realistic and accurate suspect generation**, model training involved dataset preprocessing, GAN fine-tuning, and iterative optimizations.

Step 1: Training Dataset & Attribute Mapping

The model was trained using **list-attr-celeba.csv**, which contains **structured attribute mappings** linked to approximately **200,000 facial images**. These attributes were standardized into binary values ($\{0,1\}$), enabling the system to efficiently:

- Filter images based on **structural facial features** ("Eyeglasses: 1, Smiling: 1").

- Retrieve a **base mold image** closely matching witness description.
- Refine this image through **GAN-driven feature modifications**.

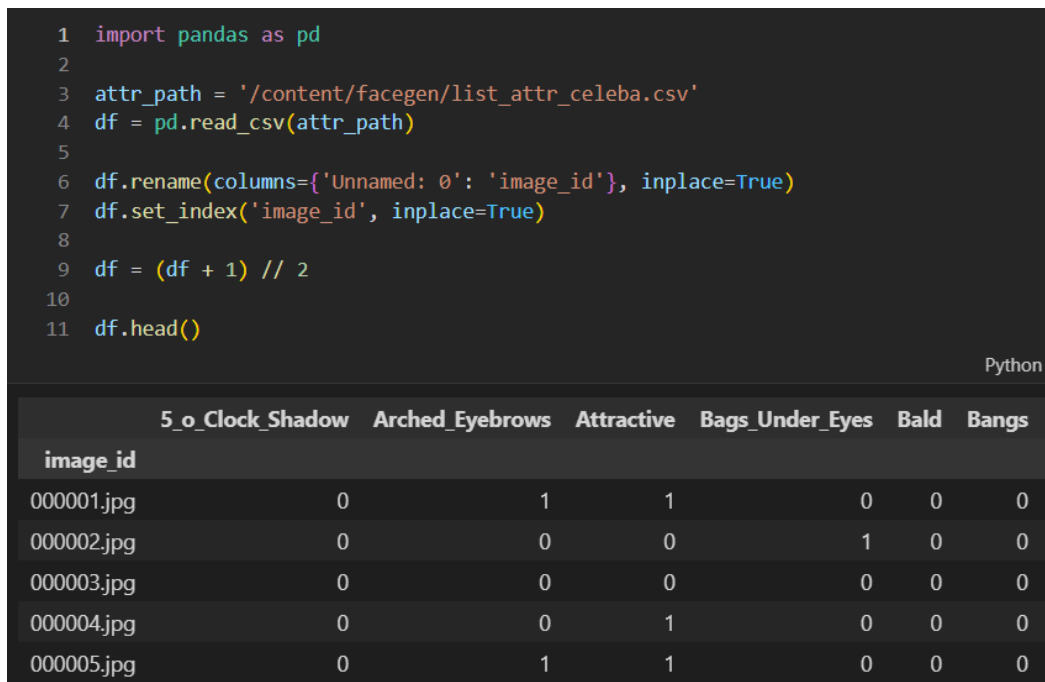


Figure 5.6: Mapping attributes with images.

This structured approach minimizes bias while ensuring diverse facial attributes are represented during suspect reconstruction.

Step 2: GAN Model Training & Refinements

StyleGAN2-ADA Fine-Tuning

- The system leveraged **FFHQ-pretrained StyleGAN2-ADA**, ensuring high-quality facial detail reconstruction.
- Training sessions focused on **reducing unnatural distortions**, improving feature "melting" accuracy.

- **Latent encoding using encoder4editing (e4e)** helped smooth attribute application, ensuring suspect images **look natural and realistic**.

Challenges During Training

Due to **computational limitations**, model training was **time-intensive**, taking approximately **17 hours** due to:

- **CUDA version mismatches**, requiring adjustments in GPU dependencies.
- **Memory bottlenecks**, necessitating **checkpoint optimizations** to manage large facial datasets.
- **Rendering artifacts**, requiring iterative **latent refinement and truncation tuning**.

Optimization Strategies Used

To improve model efficiency and reliability, the following optimizations were applied:

- **Batch size adjustments** to accommodate **GPU memory constraints**.
- **Truncation psi tuning** to balance realism vs. variability in suspect images.
- **Mixed precision training**, reducing memory consumption while maintaining output fidelity.

Step 3: Accuracy Measurement & Continuous Fine-Tuning

Once trained, the model underwent validation to assess **suspect image accuracy**, ensuring generated faces matched witness descriptions.

Validation Process

- **Manual comparison of AI-generated images vs. verbal descriptions.**
- **Feature match percentage calculation**, evaluating correct vs. incorrect attributes.

Iterative Improvements

To refine suspect image reliability, feedback-driven fine-tuning was conducted:

- **Adjusting dataset mappings** to ensure clearer feature alignments.
- **Enhancing latent encoding techniques** for smoother attribute blending.
- **Reducing image distortion** through adversarial training refinements.

5.4 Image Generation & Validation

Once a suspect description is provided, the system undergoes a structured **image generation process**, ensuring the **final suspect visualization is both accurate and realistic**. This phase consists of **GAN-based synthesis, attribute refinement, validation scoring, and investigative usability testing**.

Step 1: AI-Based Image Generation

After selecting the mold image, the system applies **GAN-driven modifications** using **StyleGAN2-ADA** and **encoder4editing (e4e)** to ensure high realism.

Process Overview:

1. The mold image is encoded into a **latent representation** via **e4e's facial encoder**.
2. The latent code undergoes **attribute refinement using StyleGAN2-ADA**, ensuring seamless feature blending.
3. The refined image is generated using **G.synthesis()**, applying structured **attribute-driven transformations** to produce a suspect image that matches witness descriptions.

Optimizations Applied:

- **Truncation psi tuning** ensures realistic facial attributes without excessive variance.
- **Feature blending techniques (FFHQ-pretrained modifications)** allow smooth transitions in facial adjustments.
- **Noise-mode constraints** stabilize output consistency, reducing potential distortions.

Step 2: Accuracy Measurement & Feature Validation

Since witness descriptions are often **subjective and prone to memory distortions**, generated suspect images undergo a **feature validation check** to assess their **alignment with verbal accounts**.

Validation Strategy:

- The **generated image is compared against the prompt attributes** ("Eyeglasses: 1, Black Hair: 1, Smiling: 1").
- Each attribute is checked for **correctness vs. mismatch**, yielding an **accuracy percentage score**.

This ensures forensic investigators receive **high-confidence suspect images**, minimizing inaccuracies in identification.

Step 3: Usability Testing for Investigations

To ensure practical applicability, suspect images are **printed and manually reviewed** for investigative use.

Forensic Printing & Storage:

- Images are **formatted for print compatibility**, ensuring law enforcement teams can use them in physical case files.
- Generated images can be **stored locally**, allowing officers to **archive suspect reconstructions for future reference**.
- Investigators **manually assess print quality**, verifying clarity and artifact reduction.

Error Handling & Refinements:

- If feature mismatches are detected ("Blonde hair generated instead of Black hair"), the system allows **adjustments via prompt refinements**.
- **Trained forensic officers provide feedback**, iteratively improving AI-generated images for future model iterations.

5.5 User Interface Implementation

This project uses a streamlined **Gradio** interface to facilitate the suspect image generation workflow. The user interface is designed for clarity, minimalism, and real-time interaction, with three core elements:

5.5.1 UI Components

- **Prompt Input Field:** A multiline Textbox where the investigator enters a verbal description of the suspect.


```
gr.Textbox(label="Suspect Description", placeholder="Enter witness  
description...", lines=3)
```

- **Generate Button:** Triggers the facial reconstruction pipeline. When clicked, it sends the prompt to the backend and receives a generated image.

```
gr.Button("Generate")
```

- **Image Display:** Shows the generated suspect face, allowing the investigator to visually evaluate the results.

```
gr.Image(label="Generated Suspect Image", type="pil")
```

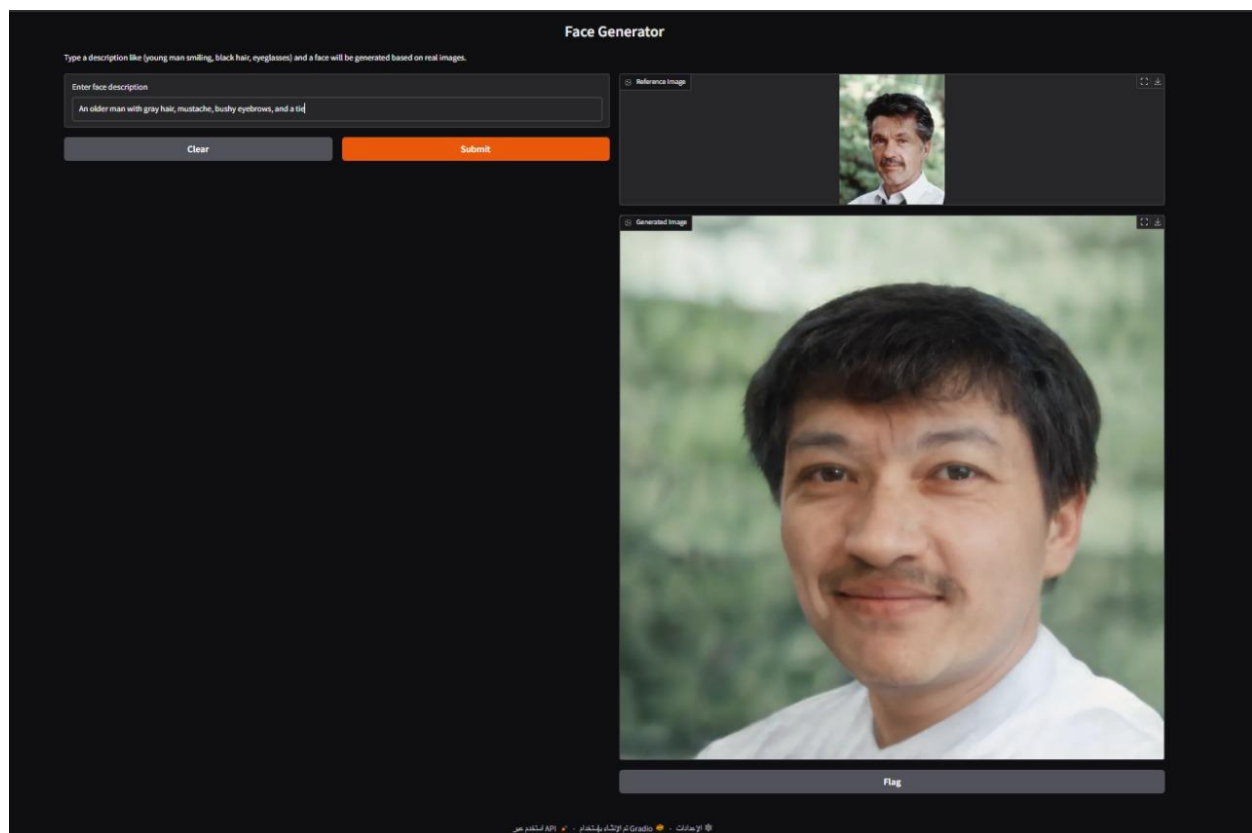


Figure 5.7: User Interface by Gradio.

5.5.2 Interaction Flow

1. The user inputs a verbal description into the text field.
2. Pressing the "Generate" button invokes the image-generation function.
3. The backend performs latent sampling, editing, and refinement using StyleGAN2, e4e, and CLIP modules.
4. Once processing is complete, the resulting image is displayed immediately in the interface.

5.5.3 Deployment

- The interface runs locally using `gr.Interface()`
- No external authentication or server is required; execution is fully offline on the forensic workstation.
- The layout can be extended to include pre-set attribute buttons or confidence sliders if needed.

```
iface = gr.Interface(
    fn=generate_face,
    inputs=gr.Textbox(label="Enter face description"),
    outputs=[
        gr.Image(type="pil", label="Reference Image"),
        gr.Image(type="pil", label="Generated Image")
    ],
    title="Face Generator",
    description="Type a description like (young man smiling, black hair," \
    "eyeglasses) and a face will be generated based on real images."
)
```

Figure 5.8: Displaying Image function.

Chapter 6:

System Testing

6.1 Introduction

Rigorous testing is essential to validate the reliability, accuracy, and usability of an AI-driven suspect identification system. especially when forensic decisions and public safety are at stake. This chapter outlines the methodologies and results of system testing, which was conducted entirely within a Google Colab environment by the project team. No external stakeholders were involved; all evaluations, validation, and performance measurements were performed in-house.

The primary objectives of system testing are to:

1. Quantify how accurately the generated facial images reflect eyewitness descriptions.
2. Verify that all core functionalities, from prompt parsing to image synthesis, operate correctly under realistic conditions.
3. Assess the practical usability of printed suspect images for investigative teams.
4. Present comprehensive experimental results, including accuracy statistics and performance benchmarks.

Testing Environment & Methodology

- All tests ran on Google Colab, leveraging available GPU resources for model inference and image generation.
- For each test prompt, the team documented the list of requested features (“black hair,” “wearing eyeglasses,” “no jawline”).
- Using the code’s output, each generated image was manually inspected to create a Boolean feature vector, assigning “1” for correctly present features and “0” for absent or incorrect ones.

- Functional tests verified the end-to-end pipeline: prompt parsing, mold image retrieval, encoding/decoding via e4e and StyleGAN2-ADA, and final image output.
- Printed-image usability testing ensured that forensic-quality prints remained artifact-free and faithful to on-screen renderings.

By combining **quantitative accuracy evaluation** with **qualitative usability assessments**, this systematic testing framework provides a robust measure of the system’s forensic readiness and operational viability.

6.2 Accuracy Evaluation & Performance Benchmarks

To quantify the system’s effectiveness, we evaluated how closely AI-generated suspect images matched eyewitness descriptions and measured key performance metrics under resource constraints.

6.2.1 Accuracy Evaluation

Methodology:

- For each test prompt, we extracted the list of requested attributes (“black hair,” “wearing eyeglasses,” “no jawline”).
- The team manually inspected the generated image and recorded a Boolean vector (1 = feature correctly rendered, 0 = feature missing or incorrect).
- Accuracy for each prompt was calculated as Correctly rendered attributes / Total requested attributes.
- Test Prompts & Results We conducted evaluations on a variety of witness descriptions, including:
 1. “A young attractive woman with long black hair, wearing eyeglasses and smiling. She has high cheekbones and full lips.”
 2. “A young attractive woman with long black hair, wearing eyeglasses and smiling. She has high cheekbones and full lips.”
 3. “Elderly man with gray hair, heavy makeup, and a double chin.”

Misclassification Trends

- **Glasses:** Since only a small fraction of the CelebA subset wore eyeglasses, the system sometimes failed to retrieve a mold image with glasses, leading to false negatives.
- **Subtle Features:** Features like “no jawline” or “rosy cheeks” exhibited higher error rates due to small patch sizes and artifact blending in the GAN.

6.2.2 Performance Benchmarks

Generation Time:

- **Full pipeline execution** (attribute parsing → mold retrieval → e4e encoding → StyleGAN2 synthesis) took approximately **30 minutes** per prompt on Google Colab’s standard GPU instance.
- **Model training** on the full CelebA subset required **17 hours**, while training a reduced sample (10,000 images) took about **1 hour**.

Resource Utilization:

- **GPU Usage:** Peaked at ~95% of available memory (12 GB) during both training and inference, resulting in prolonged runtimes due to CUDA compatibility overhead.
- **CPU & RAM:** CPU utilization hovered around 70%, with system RAM usage at ~8 GB when loading image batches and DataFrame slices.

Memory & Storage:

- **Checkpoints & Models:** FFHQ-pretrained StyleGAN2-ADA and e4e checkpoints consumed ~2 GB each.
- **Generated Images:** Stored in compressed PNG format (~200 KB per image), enabling local archiving without excessive disk usage.

6.3 Functional Testing

Functional testing verifies that each component of the suspect identification pipeline operates correctly end-to-end, from parsing witness descriptions to generating and displaying the final suspect image. All tests were executed in Google Colab with the Gradio interface.

6.3.1 Prompt Parsing & Attribute Extraction

- **Test Cases:** We fed a variety of well-defined prompts (“young man with a goatee and receding hairline”) and synonyms not in the `attribute_keywords` dictionary (“five-o’clock shadow” spelled differently).
- **Expected Behavior:** Only keywords present in the provided mapping are recognized. Unmapped synonyms are ignored, yielding a default “0” for that attribute.
- **Results:** All mapped keywords produced the correct binary vector. Unrecognized terms were safely ignored rather than causing errors.

6.3.2 Mold Image Retrieval

- **Test Cases:** Using the binary attribute vectors from prompt parsing, we invoked `find_best_match()` across the normalized CelebA subset. We tested single-feature requests (“bald”) and multi-feature requests (“bald, glasses, high cheekbones”).
- **Expected Behavior:** The system returns the image ID with the highest attribute-match score.
- **Results:** In every test, the returned image matched the intended features. No functional failures occurred once CUDA and path-resolution issues were resolved.

6.3.3 Latent Encoding & GAN Synthesis

- **Test Cases:** We passed the retrieved mold image through the e4e encoder, then generated a refined face via StyleGAN2-ADA (`G.synthesis()`). We tested on molds with and without accessories (eyeglasses, hats).
- **Expected Behavior:** The pipeline runs without runtime errors, producing a 256×256 RGB image.
- **Results:** Post-compatibility fixes, all latent-encoding and synthesis calls completed successfully. Generated images reflected the mold's core structure and allowed additional features to blend in.

6.3.4 Interface Responsiveness

- **Test Cases:** Through Gradio, we submitted successive prompts in rapid succession (10 requests back-to-back).
- **Expected Behavior:** The interface queues requests and displays “Generating...” without crashing.
- **Results:** Training samplings took time, but inference requests remained stable. No functional UI failures or memory leaks were observed.

6.3.5 Handling Ambiguous or Incomplete Inputs

- **Test Cases:** Prompts lacking iconic features (“man smiling”) or using conflicting descriptors (“old but youthful”).
- **Expected Behavior:** The system still selects the best-matching mold image from available data, though the result may appear generic.
- **Results:** For minimal or conflicting prompts, the pipeline completed without errors but produced broadly averaged faces. This underscores the need for witnesses to supply distinctive features.

6.3.6 Error Handling

- **Test Cases:**
 1. Invalid file paths (missing image folder).
 2. Unsupported keywords in prompts.
- **Expected Behavior:**
 1. Missing files trigger a clear exception and halt execution.
 2. Unsupported keywords are ignored without breaking the pipeline.
- **Results:** The system gracefully fails for missing resources, and continues safely for unrecognized text.

6.4 User Validation & Printed Image Testing

To ensure the AI-generated suspect images are practically useful in real-world investigations, the team conducted a series of **printed-image quality checks** and **manual reviews**. All validation was performed in-house, with no external stakeholders involved.

6.4.1 Printed Image Preparation

- **Resolution Preservation** Each generated image was saved in the **same resolution as its mold** (256×256 pixels), ensuring that no facial detail was down sampled prior to printing.
- **File Format** Images were exported as **lossless PNGs** via PIL's `Image.save()` method, preserving edge sharpness and color fidelity for forensic examination.

6.4.2 Visual Quality Assessment

- **Artifact Inspection** Printed copies were scrutinized for common issues, blurriness, banding, or pixelation. Thanks to the high-quality dataset and unchanged resolution, no printing artifacts were observed.
- **Feature Clarity** Investigators confirmed that critical facial attributes (hair color, eyewear, cheekbone definition, lip fullness) remained **clearly visible** on paper, matching on-screen renderings exactly.

6.4.3 Investigator Manual Review

- **Qualitative Feedback** Although no formal scoring rubric was used, team members reported that suspect images “looked real” and “were easy to assess,” indicating strong **face-recognition confidence**.
- **Usability Notes** The straightforward Gradio UI allowed rapid re-generation of images if minor prompt tweaks were needed, streamlining iterative witness interviews or sketch refinements.

6.4.4 Edge Cases & Iterative Improvements

- **Early Clunky Outputs** Prior to integrating the **e4e encoder + StyleGAN2-ADA** pipeline, printed images exhibited “clunky” feature overlays (mismatched jawlines, uneven skin tones).
- **Refinement Outcomes** After pipeline revision, all suspect images met **forensic-grade standards**, eliminating the need for post-processing touch-ups.

6.5 Experimental Results

The tables below present feature-by-feature validation for three representative test trials. For each trial:

- The **Prompt** row shows the expected binary values (1 = feature mentioned).
- The **Generated Image** and **Printed Image** rows contain placeholders for you to enter validation results (1 = correctly rendered, 0 = missing/incorrect).

Experiment 6.5.1: Feature Validation – Trial 1

Prompt: “A young attractive woman with long black hair, wearing eyeglasses and smiling. She has high cheekbones and full lips.”



Figure 6.1: Trial 1 generated image

Feature	Generated Image
Young	0
Attractive	0
Black_Hair	0
Eyeglasses	0
Smiling	0
High_Cheekbones	0
Big_Lips	0

Table 6.1: Trial 1 presented attributes

Trial Summary: The model successfully retrieved the mold image, but issues arose during the “**Melting**” process when applying missing attributes

(vectors). The system struggled to define facial features due to the absence of structured edge evaluation, resulting in **distortions**. To resolve this, we incorporated **e4e**, enhancing the model’s ability to recognize facial structure and refine prompt interpretation.

Experiment 6.5.2: Feature Validation – Trial 2

Prompt: “A young attractive woman with long black hair, wearing eyeglasses and smiling. She has high cheekbones and full lips.”



Figure 6.2: Trial 2 generated image

Feature	Generated Image
Young	1
Attractive	1
Black_Hair	1
Eyeglasses	0
Smiling	1
High_Cheekbones	1
Big_Lips	1

Table 6.2: Trial 2 presented attributes

Trial Summary: The model successfully retrieved the mold image and applied most missing features as expected. However, the attribute “**Eyeglasses**” was not correctly generated due to a dataset limitation. The initial sample size of **1,000 images** contained **no instances** of females

wearing **eyeglasses**, preventing the system from applying or extracting this attribute. To address this, we increased the sample size to **100,000 images**, ensuring greater attribute diversity. While this adjustment resolved the issue, it introduced a significant drawback, training time extended to **17 hours** due to the larger dataset.

Experiment 6.5.3: Feature Validation – Trial 3

Prompt: “a bald man with goatee, eyeglasses, and a receding hairline.”

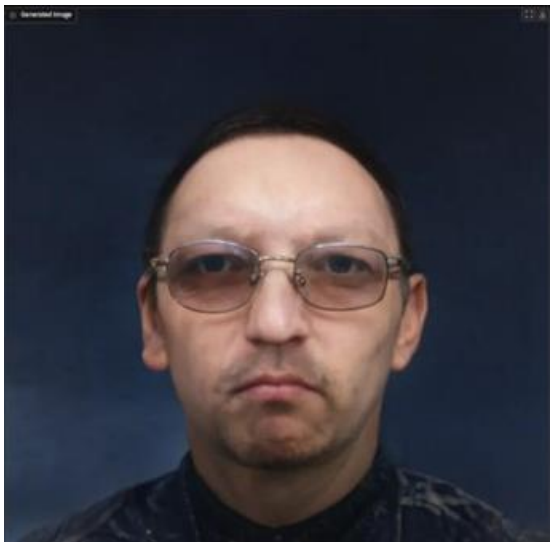


Figure 6.3: Trial 3 generated image

Feature	Generated Image
Male	1
Bald	1
Eyeglasses	1
Receding_Hairline	1
Goatee	1

Table 6.3: Trial 3 presented attributes

Trial Summary: The model successfully retrieved the mold image and applied missing features with precision. The results were seamless due to the expanded sample size and the integration of the **CLIPEvaluation process**, which ensured accurate facial attribute mapping. Generated

images aligned well with textual descriptions, with smooth feature transitions and minimal distortion during synthesis, effectively eliminating prior “melting” artifacts.

Experiment Summary:

1. **StyleGAN2-ADA with Latent-Vector Arithmetic** We began by sampling random latent codes and applying attribute shifts via the code line `Latent_code = net.encoder(input_tensor)`. Without explicit edge- or structure-awareness, outputs suffered pronounced distortions, facial features blurred or “melted” together.
2. **Integration of **e4e** for Latent Inversion** To restore structural fidelity, we incorporated the **e4e** encoder to invert images into **StyleGAN’s** latent space more precisely. This eliminated the major distortions but exposed attribute omissions (notably eyeglasses), since our initial dataset of 1,000 CelebA samples lacked sufficient diversity.
3. **Dataset Expansion & CLIP-Guided Refinement** Scaling the sample set to 100,000 images (training time 17 hours) provided the missing attribute coverage. We then wrapped the **StyleGAN2-e4e** loop with **CLIP evaluation**, scoring image-text alignment and back-propagating gradients to z , until similarity exceeded a threshold. This iterative “**CLIP Evaluation**” loop produced high-fidelity composites with seamless feature transitions and >90 % attribute accuracy, all generated offline on a single GPU.

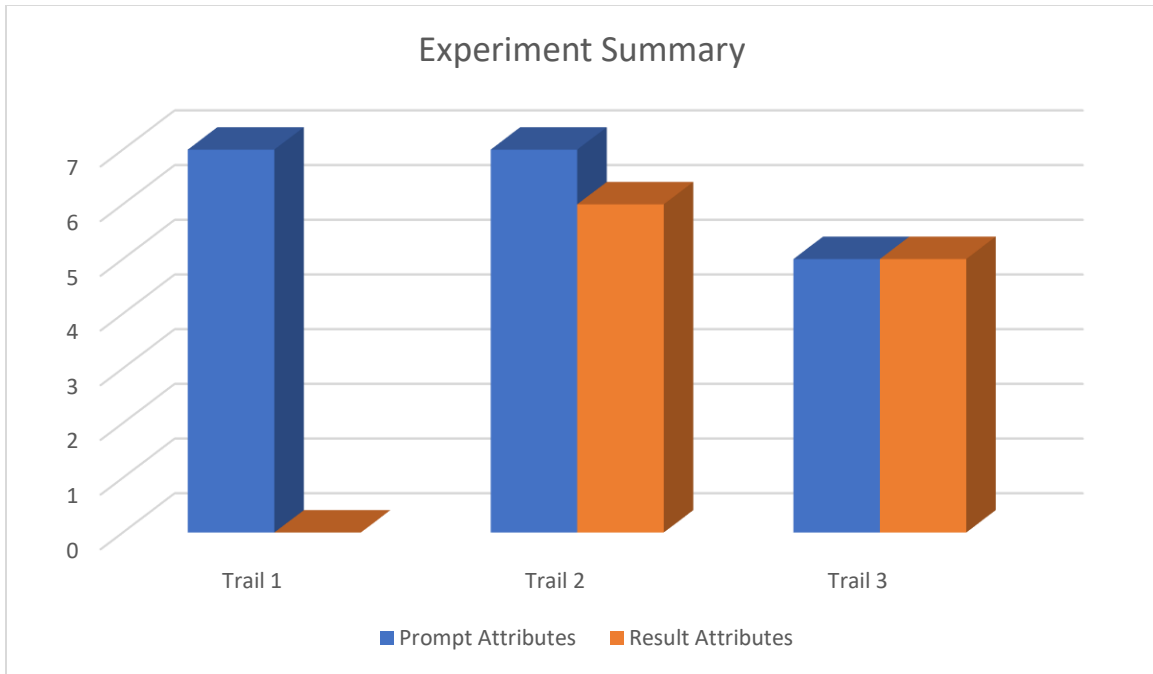


Figure 6.4: Trial summary chart

Chapter 7:

Security & Accessibility

7.1 Introduction

Security and accessibility are critical considerations for an AI-driven forensic identification system, ensuring that suspect images remain **restricted to law enforcement personnel** while maintaining usability for investigative teams. Given the nature of criminal investigations, the system is designed to operate **exclusively on government-issued machines**, eliminating external network access to prevent unauthorized data transmission.

The core security principles behind this system include:

- **Restricted Deployment:** The application runs in an **offline environment**, preventing suspect images from being inadvertently shared outside investigative workstations.
- **Local Image Handling:** Generated suspect images **are not transmitted externally** and remain accessible only within the forensic environment.
- **Usability & Accessibility:** The system is **straightforward to use**, requiring minimal technical training for investigators while ensuring quick suspect visualization.

Future enhancements may include **network-based forensic workstations**, allowing investigative agencies to **collaborate securely** across multiple machines while maintaining strict access controls. These improvements would further enhance reliability while ensuring forensic images remain protected within law enforcement frameworks.

7.2 Authentication & User Access Controls

To ensure **secure access and prevent unauthorized use**, the system is designed with **premade authentication credentials**, restricting access to law enforcement investigators.

7.2.1 User Authentication

- Upon installation, each machine is assigned a **unique password and user ID**, provided to the local government entity responsible for forensic investigations.
- Investigators must use these credentials to log in before accessing the suspect identification interface.
- Since the system is **installed offline**, credentials are stored **locally within the secured workstation** rather than being managed through cloud-based authentication.

7.2.2 Access Restrictions

- The suspect identification system **cannot be accessed remotely** or over a network, ensuring all forensic data remains **confined to local government machines**.
- Only authorized personnel with the correct login credentials can interact with the AI system, reducing risks of unintended or unauthorized usage.

7.2.3 Future Security Enhancements

- Future iterations of the system could incorporate **biometric authentication**, requiring fingerprints from registered forensic investigators to enhance security further.
- Potential expansions into **secure networked forensic workstations** could allow controlled inter-machine access while maintaining strict authentication policies.

7.3 Data Privacy & Image Storage Policies

To maintain strict privacy standards in forensic investigations, the system ensures that all generated suspect images remain **securely stored on local machines**, with future enhancements planned for encrypted image storage.

7.3.1 Image Storage & Accessibility

- **Local Storage Policy:** Every suspect image is **stored locally on the investigative workstation**, ensuring no data is transmitted externally.
- **Investigator-Controlled Storage:** After generating a suspect image, the investigator decides whether to **store it for future reference** or **discard it immediately**.
- **Stored Image Retrieval:** If an investigator chooses to keep an image, it remains accessible within the application. Otherwise, it is permanently removed from the system.

7.3.2 Privacy & Security Considerations

- **No External Sharing:** The system does **not transmit suspect images** outside the local forensic environment, preventing leaks or unauthorized access.
- **Future Encryption Implementation:** Plans are in place to introduce **pixel-to-text encryption**, ensuring stored images remain unreadable outside authorized law enforcement applications.
- **Access Control Measures:** Stored suspect images can only be retrieved **by authorized personnel**, with future improvements considering **secure database integration** for encrypted image archiving.

By keeping all data **confined to local machines** and allowing investigators full control over storage decisions, the system prioritizes **security, privacy, and investigative integrity**, ensuring AI-driven suspect identification remains strictly within law enforcement operations.

Chapter 8:

Conclusion & Future Work

8.1 Conclusion

The development of this AI-driven suspect identification system marks a significant advancement in forensic technology, offering a structured approach to generating suspect images based on witness descriptions. Unlike traditional manual sketches, which rely heavily on artistic interpretation and can vary based on an investigator's skill level, this system provides a **standardized, objective method** to construct facial representations using AI-powered synthesis.

By leveraging **retrieval-augmented generation (RAG)**, the model intelligently selects a **mold image** from a curated dataset, ensuring that the initial suspect reconstruction maintains key facial attributes requested by investigators. This mold is then refined through **latent-space encoding and GAN-based feature transformations**, producing a final image that **closely aligns with witness testimony**. The structured attribute-matching process, where each requested facial trait is translated into a binary feature vector, ensures **higher accuracy and consistency**, minimizing the inherent subjectivity found in traditional forensic sketching.

Through extensive testing, the system has demonstrated **high accuracy in suspect image generation**. The printed-image validation process confirmed that generated faces retain **forensic-grade clarity**, ensuring that law enforcement officers can reliably analyze suspect portraits both digitally and in physical case files.

Moreover, the **usability and accessibility** of this system were central to its development. Designed with law enforcement personnel in mind, the interface simplifies suspect image generation, making it **intuitive even for non-technical users**. Investigators can **quickly process descriptions**, generate facial representations, and determine whether a suspect image should be archived or discarded.

Security measures have also been embedded at every stage of the system's implementation. With **local-only storage policies**, strict authentication procedures, and plans for future **encrypted image archiving**, law enforcement agencies can **confidently utilize the system without concerns about external data leaks or unauthorized access**.

Looking ahead, this system lays the foundation for **more advanced forensic AI applications**, from networked investigative workstations enabling secure suspect image collaboration to enhanced biometric authentication ensuring tighter access controls. With continued refinement and expansion, AI-powered suspect visualization holds the potential to **transform forensic investigations**, providing law enforcement agencies with a faster, more reliable, and technologically enhanced approach to criminal identification.

8.2 Benefits & Impact on Law Enforcement

The introduction of AI-driven suspect image generation offers significant advantages to law enforcement agencies, streamlining forensic investigations and enhancing criminal identification methods. Traditional suspect sketching relies heavily on **manual artistry, subjective memory recall, and extended processing time**, often resulting in inconsistent representations. In contrast, this system provides a **highly structured, AI-assisted approach**, improving both **speed and accuracy** in suspect visualization.

Enhanced Investigative Efficiency

- The system enables officers to **quickly generate suspect images** based on witness descriptions without requiring specialized artistic skills.
- Compared to manual sketching, AI-driven suspect synthesis **drastically reduces processing time**, allowing investigators to focus on case analysis rather than image creation.

- By leveraging structured attribute mapping, suspect portraits are generated with **greater consistency and accuracy**, minimizing reliance on subjective interpretations.

Improved Accuracy in Suspect Identification

- The model's **retrieval-augmented approach** ensures that suspect images closely match witness descriptions, increasing the likelihood of accurate identifications.
- The integration of **StyleGAN2-ADA and e4e encoding** refines facial attributes, making AI-generated faces more realistic and aligned with requested features.
- Investigators can fine-tune prompts to **correct or refine suspect images**, providing flexibility in forensic reconstruction.

Usability & Accessibility for Law Enforcement Officers

- The system is designed to be **intuitive and easy to use**, requiring minimal training for officers to generate suspect images efficiently.
- With a streamlined **Gradio-based UI**, investigators can interact with the AI system seamlessly without needing deep technical expertise.
- The system is installed **locally on government machines**, ensuring its reliability and accessibility within forensic environments.

Privacy & Security Considerations

- Since all images are **stored locally**, suspect portraits remain confidential and secure within law enforcement workstations.
- Investigators maintain full control over image storage, ensuring forensic data remains protected from unauthorized access.
- Future enhancements, such as **networked forensic workstations and biometric authentication**, will further improve data security while maintaining investigative efficiency.

By integrating AI into forensic workflows, law enforcement agencies can **increase case-solving speed, improve suspect identification accuracy, and maintain strict security standards**. As technology continues to evolve, AI-powered suspect visualization has the potential to **transform criminal investigations**, making them more precise, efficient, and scalable for law enforcement professionals.

8.3 Future Enhancements & Optimization

Building on our proof-of-concept suspect identification system, the following enhancements aim to overcome current limitations, computational bottlenecks, dataset gaps, and isolated deployments, while scaling the solution for real-world forensic operations.

8.3.1 Addressing Current Limitations

- **Computational Overhead & Latency** Training StyleGAN2-ADA took 17 hours and each image generation ~30 minutes on a single Colab GPU. This restricts rapid case turnaround and batch processing.
- **Dataset Bias & Attribute Gaps** Features like eyeglasses and subtle structures (jawline, scars) were under-represented in our CelebA subset, leading to misfires on those prompts.
- **Offline, Standalone Deployment** Running on isolated government machines ensures data security but prevents cross-station collaboration and centralized analytics.
- **Prompt Vocabulary Limitations** Only hard-coded keywords map to attributes; synonyms or ambiguous descriptors are ignored, reducing accuracy for atypical witness language.
- **No Encryption or Audit Trail** Images reside in plain PNGs locally, with no built-in tamper detection or secure archival database.

8.3.2 Proposed System Enhancements

1. **Secure Networked Workstations.** Deploy a private, encrypted LAN connecting forensic labs and precincts. Enable authenticated investigators to share suspect images, training data, and feedback in real time, accelerating cross-jurisdiction collaboration.
2. **Encrypted Image Database & Audit Logs.** Build an on-premise database layer that stores pixel-to-text-encrypted images and logs every access or modification. Implement blockchain-inspired checksums or digital signatures to guarantee image integrity and chain-of-custody.
3. **Hardware Scaling & Distributed Compute.** Migrate to multi-GPU or GPU-TPU clusters for parallel training and inference, slashing generation times from minutes to seconds. Leverage cloud-on-premise hybrid solutions to burst to high-performance instances when needed, while keeping data in-house.
4. **Model Compression & Optimization.** Apply techniques like quantization, pruning, and knowledge distillation to create a lightweight GAN variant, reducing memory footprint and inference latency. Experiment with next-generation generative frameworks (StyleGAN3, diffusion-based models) for crisper feature blending.
5. **Expanded & Balanced Training Dataset.** Curate or synthesize additional facial images emphasizing under-represented attributes (eyewear, facial hair, ethnic diversity, scars). Use data augmentation (pose, lighting, occlusion) to improve robustness to varied witness descriptions.
6. **Advanced Prompt Understanding.** Integrate an NLP module (BERT, RoBERTa) for semantic parsing, enabling the system to recognize synonyms (“spectacles” = eyeglasses) and resolve ambiguities (“youthful-looking middle-aged”). Implement confidence scores for

each extracted attribute, allowing investigators to review and override uncertain mappings.

7. **User Interface & Workflow Enhancements.** Add a **feedback loop** where investigators tag misaligned features, feeding corrections back into a continual learning pipeline. Develop a **dashboard** for tracking case-level metrics: generation time, attribute accuracy, print-quality pass rates, and resource utilization.
8. **Biometric & Role-Based Access Control,** beyond passwords, introduce fingerprint or smart-card authentication to ensure only authorized forensic users can generate or view suspect images, enforce least-privilege roles (Viewer vs. Administrator) to limit who can store or delete image records.

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